

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-Q

(Mark One)



QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the quarterly period ended September 30, 2019

Or



TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Commission File Number 001-36198

INTERCONTINENTAL EXCHANGE, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

5660 New Northside Drive, 30328
Atlanta, Georgia (Zip Code)

(Address of principal executive offices)

46-2286804

(IRS Employer
Identification Number)

(770) 857-4700

Registrant's telephone number, including area code

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, \$0.01 par value per share	ICE	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Exchange Act during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of October 28, 2019, the number of shares of the registrant's Common Stock outstanding was 556,849,511 shares.

INTERCONTINENTAL EXCHANGE, INC.
Form 10-Q
Quarterly Period Ended September 30, 2019
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PART I. Financial Statements

Item 1. Consolidated Financial Statements (Unaudited)

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Balance Sheets
(In millions, except per share amounts)
(Unaudited)

	As of September 30, 2019	As of December 31, 2018
Assets:		
Current assets:		
Cash and cash equivalents	\$ 655	\$ 724
Short-term restricted cash and cash equivalents	945	818
Customer accounts receivable, net of allowance for doubtful accounts of \$8 and \$7 at September 30, 2019 and December 31, 2018, respectively	1,056	953
Margin deposits, guaranty funds and delivery contracts receivable	65,988	63,955
Prepaid expenses and other current assets	227	242
Total current assets	68,871	66,692
Property and equipment, net	1,505	1,241
Other non-current assets:		
Goodwill	13,310	13,085
Other intangible assets, net	10,330	10,462
Long-term restricted cash and cash equivalents	389	330
Other non-current assets	934	981
Total other non-current assets	24,963	24,858
Total assets	\$ 95,339	\$ 92,791
Liabilities and Equity:		
Current liabilities:		
Accounts payable and accrued liabilities	\$ 543	\$ 521
Section 31 fees payable	34	105
Accrued salaries and benefits	228	280
Deferred revenue	247	135
Short-term debt	1,329	951
Margin deposits, guaranty funds and delivery contracts payable	65,988	63,955
Other current liabilities	165	161
Total current liabilities	68,534	66,108
Non-current liabilities:		
Non-current deferred tax liability, net	2,281	2,337
Long-term debt	6,496	6,490
Accrued employee benefits	197	204
Non-current operating lease liability	287	—
Other non-current liabilities	287	350
Total non-current liabilities	9,548	9,381
Total liabilities	78,082	75,489
Commitments and contingencies		
Redeemable non-controlling interest in consolidated subsidiaries	72	71
Equity:		
Intercontinental Exchange, Inc. stockholders' equity:		
Preferred stock, \$0.01 par value; 100 shares authorized; no shares issued or outstanding at September 30, 2019 and December 31, 2018	—	—

Common stock, \$0.01 par value; 1,500 shares authorized; 607 and 604 shares issued at September 30, 2019 and December 31, 2018, respectively, and 558 and 569 shares outstanding at September 30, 2019 and December 31, 2018, respectively	6	6
Treasury stock, at cost; 49 and 35 shares at September 30, 2019 and December 31, 2018, respectively	(3,538)	(2,354)
Additional paid-in capital	11,706	11,547
Retained earnings	9,335	8,317
Accumulated other comprehensive loss	(348)	(315)
Total Intercontinental Exchange, Inc. stockholders' equity	17,161	17,201
Non-controlling interest in consolidated subsidiaries	24	30
Total equity	17,185	17,231
Total liabilities and equity	<u>\$ 95,339</u>	<u>\$ 92,791</u>

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Income
(In millions, except per share amounts)
(Unaudited)

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2019	2018	2019	2018
Revenues:				
Transaction and clearing, net	\$ 2,698	\$ 2,522	\$ 929	\$ 760
Data services	1,652	1,576	553	530
Listings	336	332	114	112
Other revenues	194	169	67	61
Total revenues	4,880	4,599	1,663	1,463
Transaction-based expenses:				
Section 31 fees	274	272	105	61
Cash liquidity payments, routing and clearing	702	656	222	202
Total revenues, less transaction-based expenses	3,904	3,671	1,336	1,200
Operating expenses:				
Compensation and benefits	768	732	261	251
Professional services	97	91	35	32
Acquisition-related transaction and integration costs	1	33	—	6
Technology and communication	346	320	126	107
Rent and occupancy	52	50	17	17
Selling, general and administrative	116	109	33	37
Depreciation and amortization	473	429	158	148
Total operating expenses	1,853	1,764	630	598
Operating income	2,051	1,907	706	602
Other income (expense):				
Interest income	27	15	8	6
Interest expense	(214)	(173)	(72)	(66)
Other income, net	30	33	(2)	12
Other income (expense), net	(157)	(125)	(66)	(48)
Income before income tax expense	1,894	1,782	640	554
Income tax expense	387	381	103	89
Net income	\$ 1,507	\$ 1,401	\$ 537	\$ 465
Net income attributable to non-controlling interest	(22)	(24)	(8)	(7)
Net income attributable to Intercontinental Exchange, Inc.	\$ 1,485	\$ 1,377	\$ 529	\$ 458
Earnings per share attributable to Intercontinental Exchange, Inc. common stockholders:				
Basic	\$ 2.64	\$ 2.39	\$ 0.95	\$ 0.80
Diluted	\$ 2.62	\$ 2.37	\$ 0.94	\$ 0.79
Weighted average common shares outstanding:				
Basic	563	577	559	572
Diluted	566	581	563	576

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Comprehensive Income
(In millions)
(Unaudited)

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2019	2018	2019	2018
Net income	\$ 1,507	\$ 1,401	\$ 537	\$ 465
Other comprehensive income (loss):				
Foreign currency translation adjustments, net of tax benefit of \$1 for both the nine months ended September 30, 2019 and 2018 and \$1 for the three months ended September 30, 2019	(32)	(51)	(39)	(9)
Change in equity method investment	(1)	—	—	—
Other comprehensive income (loss)	(33)	(51)	(39)	(9)
Comprehensive income	\$ 1,474	\$ 1,350	\$ 498	\$ 456
Comprehensive income attributable to non-controlling interest	(22)	(24)	(8)	(7)
Comprehensive income attributable to Intercontinental Exchange, Inc.	\$ 1,452	\$ 1,326	\$ 490	\$ 449

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Loss			
	Shares	Value	Shares	Value						
Balance, as of December 31, 2018	604	\$ 6	(35)	\$ (2,354)	\$ 11,547	\$ 8,317	\$ (315)	\$ 30	\$ 17,231	\$ 71
Other comprehensive loss	—	—	—	—	—	—	(33)	—	(33)	—
Exercise of common stock options	1	—	—	—	22	—	—	—	22	—
Repurchases of common stock	—	—	(14)	(1,120)	—	—	—	—	(1,120)	—
Payments relating to treasury shares	—	—	—	(64)	—	—	—	—	(64)	—
Stock-based compensation	—	—	—	—	109	—	—	—	109	3
Issuance under the employee stock purchase plan	—	—	—	—	28	—	—	—	28	—
Issuance of restricted stock	2	—	—	—	—	—	—	—	—	—
Distributions of profits	—	—	—	—	—	—	—	(29)	(29)	—
Dividends paid to stockholders	—	—	—	—	—	(467)	—	—	(467)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(22)	—	23	1	(2)
Net income	—	—	—	—	—	1,507	—	—	1,507	—
Balance, as of September 30, 2019	607	\$ 6	(49)	\$ (3,538)	\$ 11,706	\$ 9,335	\$ (348)	\$ 24	\$ 17,185	\$ 72

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Loss			
	Shares	Value	Shares	Value						
Balance, as of June 30, 2019	606	\$ 6	(45)	\$ (3,194)	\$ 11,651	\$ 8,961	\$ (309)	\$ 31	\$ 17,146	\$ 72
Other comprehensive loss	—	—	—	—	—	—	(39)	—	(39)	—
Exercise of common stock options	1	—	—	—	5	—	—	—	5	—
Repurchases of common stock	—	—	(4)	(340)	—	—	—	—	(340)	—
Payments relating to treasury shares	—	—	—	(4)	—	—	—	—	(4)	—
Stock-based compensation	—	—	—	—	36	—	—	—	36	1
Issuance under the employee stock purchase plan	—	—	—	—	14	—	—	—	14	—
Distributions of profits	—	—	—	—	—	—	—	(15)	(15)	—
Dividends paid to stockholders	—	—	—	—	—	(155)	—	—	(155)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(8)	—	8	—	(1)
Net income	—	—	—	—	—	537	—	—	537	—
Balance, as of September 30, 2019	607	\$ 6	(49)	\$ (3,538)	\$ 11,706	\$ 9,335	\$ (348)	\$ 24	\$ 17,185	\$ 72

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest — (Continued)
(In millions)
(Unaudited)

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Loss			
	Shares	Value	Shares	Value						
Balance, as of December 31, 2017	600	\$ 6	(17)	\$(1,076)	\$ 11,392	\$ 6,858	\$ (223)	\$ 28	\$ 16,985	\$ —
Other comprehensive loss	—	—	—	—	—	—	(51)	—	(51)	—
Exercise of common stock options	—	—	—	—	19	—	—	—	19	—
Repurchases of common stock	—	—	(14)	(1,059)	—	—	—	—	(1,059)	—
Payments relating to treasury shares	—	—	(1)	(78)	—	—	—	—	(78)	—
Stock-based compensation	—	—	—	—	107	—	—	—	107	—
Issuance of restricted stock	3	—	—	—	—	—	—	—	—	—
Acquisition of non-controlling interest	—	—	—	—	(23)	—	—	(2)	(25)	—
Distributions of profits	—	—	—	—	—	—	—	(28)	(28)	—
Dividends paid to stockholders	—	—	—	—	—	(417)	—	—	(417)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(24)	—	24	—	—
Net income	—	—	—	—	—	1,401	—	—	1,401	—
Balance, as of September 30, 2018	603	\$ 6	(32)	\$(2,213)	\$ 11,495	\$ 7,818	\$ (274)	\$ 22	\$ 16,854	\$ —

	Intercontinental Exchange, Inc. Stockholders' Equity							Non-Controlling Interest in Consolidated Subsidiaries	Total Equity	Redeemable Non-Controlling Interest
	Common Stock		Treasury Stock		Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Loss			
	Shares	Value	Shares	Value						
Balance, as of June 30, 2018	603	\$ 6	(29)	\$ (1,911)	\$ 11,477	\$ 7,498	\$ (265)	\$ 35	\$ 16,840	\$ —
Other comprehensive loss	—	—	—	—	—	—	(9)	—	(9)	—
Exercise of common stock options	—	—	—	—	7	—	—	—	7	—
Repurchases of common stock	—	—	(4)	(300)	—	—	—	—	(300)	—
Payments relating to treasury shares	—	—	1	(2)	—	—	—	—	(2)	—
Stock-based compensation	—	—	—	—	34	—	—	—	34	—
Acquisition of non-controlling interest	—	—	—	—	(23)	—	—	(2)	(25)	—
Distributions of profits	—	—	—	—	—	—	—	(18)	(18)	—
Dividends paid to stockholders	—	—	—	—	—	(138)	—	—	(138)	—
Net income attributable to non-controlling interest	—	—	—	—	—	(7)	—	7	—	—
Net income	—	—	—	—	—	465	—	—	465	—
Balance, as of September 30, 2018	603	\$ 6	(32)	\$ (2,213)	\$ 11,495	\$ 7,818	\$ (274)	\$ 22	\$ 16,854	\$ —

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Consolidated Statements of Cash Flows
(In millions)
(Unaudited)

	Nine Months Ended September 30,	
	2019	2018
Operating activities:		
Net income	\$ 1,507	\$ 1,401
Adjustments to reconcile net income to net cash provided by operating activities:		
Depreciation and amortization	473	429
Stock-based compensation	100	93
Deferred taxes	(52)	(2)
Other	(34)	(11)
Changes in assets and liabilities:		
Customer accounts receivable	(97)	(119)
Other current and non-current assets	22	(24)
Section 31 fees payable	(70)	(117)
Deferred revenue	114	124
Other current and non-current liabilities	(81)	(39)
Total adjustments	375	334
Net cash provided by operating activities	1,882	1,735
Investing activities:		
Capital expenditures	(87)	(49)
Capitalized software development costs	(116)	(112)
Cash paid for acquisitions	(352)	(1,151)
Return of capital from equity method investment	44	—
Purchases of equity investments	—	(305)
Other	(1)	—
Net cash used in investing activities	(512)	(1,617)
Financing activities:		
Proceeds from debt facilities	11	2,213
Repayments of debt facilities	—	(600)
Proceeds from/(repayments of) commercial paper, net	367	(35)
Repurchases of common stock	(1,120)	(1,059)
Dividends to stockholders	(467)	(417)
Payments relating to treasury shares received for restricted stock tax payments and stock option exercises	(64)	(78)
Acquisition of non-controlling interest	—	(35)
Other	22	(8)
Net cash used in financing activities	(1,251)	(19)
Effect of exchange rate changes on cash, cash equivalents and restricted cash and cash equivalents	(2)	(5)
Net increase in cash, cash equivalents, and restricted cash and cash equivalents	117	94
Cash, cash equivalents, and restricted cash and cash equivalents, beginning of period	1,872	1,568
Cash, cash equivalents, and restricted cash and cash equivalents, end of period	\$ 1,989	\$ 1,662
Supplemental cash flow disclosure:		
Cash paid for income taxes	\$ 457	\$ 424
Cash paid for interest	\$ 214	\$ 136

See accompanying notes.

Intercontinental Exchange, Inc. and Subsidiaries
Notes to Consolidated Financial Statements
(Unaudited)

1. Description of Business

Nature of Business and Organization

We are a leading global operator of regulated exchanges, clearing houses and listings venues, and a provider of data services for commodity, financial, fixed income and equity markets. We operate regulated marketplaces for listing, trading and clearing of a broad array of derivatives contracts and securities across major asset classes, including energy and agricultural commodities, metals, interest rates, equities, exchange-traded funds, or ETFs, credit derivatives, digital assets, bonds and currencies, as well as offer mortgage and technology services. We also offer comprehensive data services to support the trading, investment, risk management and connectivity needs of customers around the world and across asset classes.

Our exchanges include derivative exchanges in the United States, or U.S., United Kingdom, or U.K., European Union, or EU, Canada and Singapore, and cash equities, equity options and bond trading venues in the U.S. We also operate over-the-counter, or OTC, markets for physical energy, fixed income and credit default swaps, or CDS, trade execution. To serve global derivatives markets, we operate central counterparty clearing houses, or CCPs, in the U.S., U.K., EU, Canada and Singapore (Note 12). We offer a range of data services globally, for financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through our markets, clearing houses, listings venues and data services, we provide comprehensive solutions for our customers to manage risk and raise capital through liquid markets, benchmark products, access to capital markets and related services. Our business is conducted as two reportable business segments, our Trading and Clearing segment and our Data and Listings segment, and the majority of our identifiable assets are located in the U.S. and U.K. (Note 15).

2. Summary of Significant Accounting Policies

Basis of Presentation

The accompanying unaudited consolidated financial statements have been prepared by us in accordance with U.S. generally accepted accounting principles, or U.S. GAAP, pursuant to the rules and regulations of the Securities and Exchange Commission, or SEC, regarding interim financial reporting. Accordingly, the unaudited consolidated financial statements do not include all of the information and footnotes required by U.S. GAAP for complete financial statements and should be read in conjunction with our audited consolidated financial statements and related notes thereto for the year ended December 31, 2018. The accompanying unaudited consolidated financial statements reflect all adjustments that are, in our opinion, necessary for a fair presentation of results for the interim periods presented. We believe that these adjustments are of a normal recurring nature.

Preparing financial statements requires us to make certain estimates and assumptions that affect the amounts that are reported in our consolidated financial statements and accompanying disclosures. Actual amounts could differ from those estimates. The results of operations for the nine and three months ended September 30, 2019 are not necessarily indicative of the results to be expected for any future period or the full fiscal year.

These statements include the accounts of our wholly-owned and controlled subsidiaries. All intercompany balances and transactions between us and our wholly-owned and controlled subsidiaries have been eliminated in consolidation. For consolidated subsidiaries in which our ownership is less than 100% and for which we have control over the assets and liabilities and the management of the entity, the outside stockholders' interests are shown as non-controlling interests.

Recently Adopted Accounting Pronouncements

Standard/Description	Effective Date and Adoption Considerations	Effect on Financial Statements
ASU No. 2016-02, <i>Leases</i> . Entities are required to recognize both assets and liabilities arising from finance and operating leases, along with additional qualitative and quantitative disclosures.	We adopted ASU No. 2016-02 on January 1, 2019.	Further disclosures and details on our adoption of ASU 2016-02 are discussed below.

Accounting Pronouncements Not Yet Adopted

Standard/Description	Effective Date and Adoption Considerations	Effect on Financial Statements
ASU No. 2016-13, <i>Financial Instruments - Measurement of Credit Losses on Financial Instruments</i> . Applies to all financial instruments carried at amortized cost including held-to-maturity debt securities and trade receivables. Requires financial assets carried at amortized cost to be presented at the net amount expected to be collected and requires entities to record credit losses through an allowance for credit losses on available-for-sale debt securities.	Effective at the beginning of our first quarter of fiscal year 2020, with early adoption permitted. We do not expect to early-adopt.	We are currently evaluating this guidance to determine its impact on our consolidated financial statements, and based on our preliminary assessment, we do not expect the impact of adoption of this guidance to be material.

Adoption of ASU 2016-02, "Leases"

On January 1, 2019, we adopted ASU 2016-02, *Leases*, or ASU 2016-02. This standard requires recognition of both assets and liabilities arising from finance and operating leases, along with additional qualitative and quantitative disclosures. ASU 2016-02 requires lessees to recognize a right-of-use asset representing a right to use the underlying asset over the lease term, and a corresponding lease liability on the balance sheet. Our operating leases primarily relate to our leased office space and data center facilities, and we do not have any leases classified as finance leases.

We adopted ASU 2016-02 using the modified retrospective transition method and did not restate prior periods. Using the modified retrospective approach, we applied the provisions of ASU 2016-02 beginning in the period of adoption, and elected the package of practical expedients available to us. There was no impact to the opening balance of retained earnings as a result of a cumulative-effect adjustment on the adoption date. We elected the practical expedient to not reassess lease classifications, but alternatively to carry forward our historical classifications. In addition, we elected the practical expedient of not separating lease and non-lease components as our lease arrangements are not highly dependent on other underlying assets. Our implementation of the amended lease guidance was subject to the same internal controls over financial reporting that we apply to our consolidated financial statements.

At lease inception, we review the service arrangement and components of a contract to identify if a lease or embedded lease arrangement exists. An indicator of a contract containing a lease is when we have the right to control and use an identified asset over a period of time in exchange for consideration. Operating lease right-of-use assets and liabilities are recognized at the commencement date based on the present value of lease payments over the lease term, using our estimated incremental borrowing rate. Upon adoption of ASU 2016-02, we made the policy election to not record existing or future leases with a term of 12 months or less on the balance sheet, and to recognize lease expense on a straight-line basis over the lease term. For these leases, the impact on adoption was nominal. We have also made policy elections related to capitalization thresholds and discount rates. We have elected to use a portfolio approach in consideration of our incremental borrowing rate to our population of lease agreements. Our incremental borrowing rate was determined based on our recent debt issuances that we believe are reflective of current borrowing rates. Certain lease agreements include options to extend, renew or terminate the lease agreement. As of September 30, 2019, the weighted-average remaining lease term was 6.9 years and the weighted average discount rate was 3.6%. Our lease agreements do not contain any residual value guarantees.

Upon adoption of ASU 2016-02, we recorded \$357 million in operating lease liabilities, of which \$52 million is included in other current liabilities and \$305 million is included in non-current operating lease liability within our accompanying consolidated balance sheet. We also recorded \$308 million in operating lease right-of-use assets that are included as a component of property and equipment, net, in our balance sheet and are recorded in an amount equal to our lease liability, adjusted for any remaining unamortized lease incentives such as our deferred rent balances. As part of our adoption, we eliminated \$49 million in deferred rent liabilities, of which \$2 million had previously been included in other current liabilities and \$47 million had been included in other non-current liabilities on our balance sheet. On the date of adoption, deferred rent liabilities were reclassified and presented as a reduction to the right-of-use asset, included in property and equipment, net, on our consolidated balance sheet. Our adoption did not have an impact on our consolidated income statement.

We recognize rent expense monthly on a straight-line basis for each respective operating lease, as a reduction to the right-of-use asset. For the nine and three months ended September 30, 2019, we recognized \$29 million and \$9 million, respectively, of rent expense for office space as rent and occupancy expense and \$16 million and \$5 million, respectively, of rent expense for data center facilities as technology and communication expense within our consolidated income statement. We do not have any significant variable lease costs related to building and maintenance costs, real estate taxes, or other charges.

Details of our lease asset and liability balances are as follows (in millions):

	As of September 30, 2019	As of January 1, 2019
Right-of-use lease assets	\$292	\$308
Current operating lease liability	52	52
Non-current operating lease liability	287	305
Total operating lease liability	\$339	\$357

As of September 30, 2019, we estimate that our operating lease liability will be recognized in the following years (in millions):

Remainder of 2019	\$	14
2020		64
2021		63
2022		61
2023		44
Thereafter		137
Lease liability amounts repayable		383
Interest costs		44
Total operating lease liability	\$	339

Supplemental cash flow information and non-cash activity related to our operating leases are as follows:

	Nine Months Ended September 30, 2019	Three Months Ended September 30, 2019
Cash paid for amounts included in the measurement of operating lease liability	\$ 46	\$ 15
Right-of-use assets obtained in exchange for operating lease obligations	\$ 381	\$ 2

3. Acquisitions

On June 12, 2019, we acquired Simplifile, LC, or Simplifile, for \$338 million in cash. The cash consideration was gross of \$12 million in cash held by Simplifile on the date of acquisition. Simplifile offers an array of mortgage services, primarily serving as an electronic liaison between lenders, settlement agents and county recording offices, streamlining the local recording of residential mortgage transactions. The transaction expands the ICE Mortgage Services portfolio, which includes MERSCORP Holdings, Inc., or MERS.

The Simplifile purchase price was allocated to the preliminary net tangible and identifiable intangible assets and liabilities based on their estimated fair values as of June 12, 2019. The identifiable intangible assets acquired were \$113 million and included (i) customer relationships of \$104 million which have been assigned a useful life of 20 years (ii) developed technology

of \$7 million which has been assigned a useful life of 7 years and (iii) trade names of \$2 million which have been assigned a useful life of 5 years. The excess of the purchase price over the preliminary net tangible and identifiable intangible assets was \$218 million and was recorded as goodwill. Simplifile is included in our Trading and Clearing segment.

During the nine months ended September 30, 2019, our consolidated subsidiary Bakkt Holdings, LLC, or Bakkt, acquired two other companies which are not material to our operations.

4. Investments

Our equity investments, including our investments in Euroclear plc, or Euroclear, and Coinbase Global, Inc., among others, are subject to valuation under ASU 2016-01, *Financial Instruments- Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*. See Note 14 for a discussion of our determination of fair value of our financial instruments.

In addition, we own a 40% interest in the Options Clearing Corporation, or OCC, through a direct investment by the New York Stock Exchange, or NYSE, that we treat as an equity method investment. OCC serves as a clearing house for securities options, security futures, commodity futures and options on futures traded on various independent exchanges. OCC clears securities options traded on NYSE Arca and NYSE Amex Options, along with other non-affiliated exchanges, and is regulated by the SEC as a registered clearing agency and by the Commodity Futures Trading Commission, or CFTC, as a derivatives clearing organization. Under equity method accounting, each reporting period we adjust the carrying value of our OCC investment on our balance sheet by recognizing our pro-rata 40% share of the earnings or losses of OCC, with a corresponding adjustment in our statement of income to other income, after eliminating any intra-entity income or expenses. In addition, if and when OCC issues cash dividends to us, we deduct the amount of these dividends from the carrying amount of our investment.

OCC adopted a new capital plan during the first quarter of 2015, which raised \$150 million in equity capital from OCC's shareholders, including \$60 million contributed by us. Pursuant to the terms of the capital plan, in exchange for the contributions of equity capital from its shareholders, OCC was required, subject to determination by its board of directors and compliance with legal requirements, to pay an annual dividend to its shareholders on a pro rata basis. The dividend was intended to be equal to the amount (i) of after-tax income of OCC, in excess of the amount required to maintain its target capital requirement and satisfy other capital requirements, and (ii) remaining after refunds to its clearing members equal to 50% of distributable earnings before tax. Related to that capital plan, from 2015-2017 we received a total of \$31 million in dividends from OCC.

Subsequent to our \$60 million investment, certain industry participants appealed the SEC's approval of the OCC capital plan in the U.S. Court of Appeals, and in August 2017, the Court of Appeals remanded the capital plan to the SEC. On February 13, 2019, the SEC disapproved the OCC capital plan established in 2015. Consistent with the SEC's disapproval of the OCC capital plan, the OCC returned \$44 million of our original \$60 million contribution during the nine months ended September 30, 2019 as a result of the disapproval. The remaining \$16 million is expected to be returned during the fourth quarter of 2019 and will allow the OCC to maintain target capital requirements.

Following the SEC disapproval, the OCC also announced they will not be providing a refund to clearing members or declaring a dividend to shareholders for the year ended December 31, 2018, which resulted in higher reported OCC 2018 net income than we had estimated. Therefore, during the nine months ended September 30, 2019, we adjusted equity earnings in OCC by recording an additional \$19 million earnings in other income to reflect our share of OCC's 2018 net income. In addition, we recognized \$32 million and \$15 million during the nine and three months ended September 30, 2019, respectively, of equity earnings as our share of OCC's estimated 2019 profits, which is also included in other income.

5. Revenue Recognition

Substantially all of our revenues are considered to be revenues from contracts with customers. The related accounts receivable balances are recorded in our balance sheets as customer accounts receivable. We do not have obligations for warranties, returns or refunds to customers, other than the rebates discussed below, which are settled each period and therefore do not result in variable consideration. We do not have significant revenue recognized from performance obligations that were satisfied in prior periods, and we do not have any transaction price allocated to unsatisfied performance obligations other than in our deferred revenue. Deferred revenue represents our contract liabilities related to our annual, original and other listings revenues as well as certain data services, clearing services and other revenues. See Note 7 for our discussion of deferred revenue balances, activity, and expected timing of recognition. As permitted by U.S. GAAP, we have elected not to provide disclosures about transaction price allocated to unsatisfied performance obligations if contract durations are less than one year, or if we are not required to estimate the transaction price. For all of our contracts with customers, except for listings and certain data and clearing services, our performance obligations are short-term in nature and there is no significant variable consideration. See the bullets below for further descriptions of our revenue contracts. In addition, we have elected

the practical expedient of excluding sales taxes from transaction prices. We have assessed the costs incurred to obtain or fulfill a contract with a customer and determined them to be immaterial.

Certain judgments and estimates were used in the identification and timing of satisfaction of performance obligations and the related allocation of transaction price. We believe that these represent a faithful depiction of the transfer of services to our customers.

Our primary revenue contract classifications are described below. These categories best represent those with similar economic characteristics of the nature, amount, timing and uncertainty of revenues and cash flows.

- **Transaction and clearing, net** - Transaction and clearing revenues represent fees charged for the performance obligations of derivatives trading and clearing, cash, equity options and fixed income trading, as well as mortgage services. The derivatives trading and clearing fees contain two performance obligations: (1) trade execution/clearing novation and (2) risk management of open interest. While we allocate the transaction price between these two performance obligations, since they generally are satisfied almost simultaneously there is no significant deferral of revenue. Cash trading, equity options, mortgage services and fixed income fees contain one performance obligation related to each transaction which occurs instantaneously, and the revenue is recorded at the point in time of the transaction. Our transaction and clearing revenues are reported net of rebates, except for the NYSE transaction-based expenses. Rebates were \$653 million and \$615 million for the nine months ended September 30, 2019 and 2018, respectively, and \$218 million and \$183 million for the three months ended September 30, 2019 and 2018, respectively. Transaction and clearing fees can be variable based on trade volume discounts used in the determination of rebates, however virtually all volume discounts are calculated and recorded on a monthly basis. Transaction and clearing fees, as well as any volume discounts rebated to our customers, are calculated and billed monthly in accordance with our published fee schedules. We make liquidity payments to certain customers in our NYSE businesses and recognize those payments as a cost of revenue. In addition, we pay NYSE regulatory oversight fees to the SEC and collect equal amounts from our customers. These are also considered a cost of revenue, and both of these NYSE-related fees are included in transaction-based expenses. Transaction and clearing revenues and the related transaction-based expenses are all recognized in our Trading and Clearing segment. In certain of our revenue share arrangements with third parties we control the delivered contract; in these arrangements we are acting as a principal and the revenue is recorded gross.
- **Data services** - Data service revenues represent the following:
 - Pricing and analytics services provide global securities evaluations, reference data, market indices, risk analytics, derivatives pricing and other information designed to meet our customers' portfolio management, trading, risk management, reporting and regulatory compliance needs.
 - Exchange data and feeds services provide real-time, historical and derived pricing data, order book and transaction information related to our trading venues, as well as data from a broad array of third-party trading venues and news feeds.
 - Desktops and connectivity services provide the connection to our exchanges, clearing houses and data centers and comprise hosting, colocation, infrastructure, technology-based information platforms, workstations and connectivity solutions through the ICE Global Network.

The nature and timing of each contract type for the data services above are similar in nature. Data services revenues are primarily subscription-based, billed monthly, quarterly or annually in advance and recognized ratably over time as our performance obligations of data delivery are met consistently throughout the period. Considering these contracts primarily consist of single performance obligations with fixed prices, there is no variable consideration and no need to allocate the transaction price. In certain of our data contracts, where third parties are involved, we arrange for the third party to transfer the services to our customers; in these arrangements we are acting as an agent and revenue is recorded net. All data services fees are included in our Data and Listings segment.

- **Listings** - Listings revenues include original and annual listings fees, and other corporate action fees. Each distinct listing fee is allocated to multiple performance obligations including original and incremental listing and investor relations services, as well as a customer's material right to renew the option to list on our exchanges. In performing this allocation, the standalone selling price of the listing services is based on the original and annual listing fees and the standalone selling price of the investor relations services is based on its market value. All listings fees are billed upfront and the identified performance obligations are satisfied over time. Revenue related to the investor relations performance obligation is recognized ratably over a two-year period, with the remaining revenue recognized ratably over time as customers continue to list on our exchanges, which is generally estimated to be over a period of up to nine years for NYSE and up to five years for NYSE Arca and NYSE American. Listings fees related to other corporate actions are considered contract modifications of our listing contracts and are recognized ratably over time as customers

continue to list on our exchanges, which is generally estimated to be a period of six years for NYSE and three years for NYSE Arca and NYSE American. All listings fees are recognized in our Data and Listings segment.

- **Other revenues** - Other revenues primarily include interest income on certain clearing margin deposits, regulatory penalties and fines, fees for use of our facilities, regulatory fees charged to member organizations of our U.S. securities exchanges, designated market maker service fees, exchange membership fees and agricultural grading and certification fees. Generally, fees for other revenues contain one performance obligation. Because these contracts primarily consist of single performance obligations with fixed prices, there is no variable consideration and no need to allocate the transaction price. Services for other revenues are primarily satisfied at a point in time. Therefore, there is no need to allocate the fee and no deferral results as we have no further obligation to the customer at that time. Other revenues are recognized in our Trading and Clearing segment.

The following table depicts the disaggregation of our revenue according to business line and segment (in millions). Segment totals here are consistent with the segment totals in Note 15:

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Nine months ended September 30, 2019:			
Transaction and clearing, net	\$ 2,698	\$ —	\$ 2,698
Data services	—	1,652	1,652
Listings	—	336	336
Other revenues	194	—	194
Total revenues	2,892	1,988	4,880
Transaction-based expenses	976	—	976
Total revenues, less transaction-based expenses	\$ 1,916	\$ 1,988	\$ 3,904
Timing of Revenue Recognition			
Services transferred at a point in time	\$ 1,653	\$ —	\$ 1,653
Services transferred over time	263	1,988	2,251
Total revenues, less transaction-based expenses	\$ 1,916	\$ 1,988	\$ 3,904
	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Nine months ended September 30, 2018:			
Transaction and clearing, net	\$ 2,522	\$ —	\$ 2,522
Data services	—	1,576	1,576
Listings	—	332	332
Other revenues	169	—	169
Total revenues	2,691	1,908	4,599
Transaction-based expenses	928	—	928
Total revenues, less transaction-based expenses	\$ 1,763	\$ 1,908	\$ 3,671
Timing of Revenue Recognition			
Services transferred at a point in time	\$ 1,507	\$ —	\$ 1,507
Services transferred over time	256	1,908	2,164
Total revenues, less transaction-based expenses	\$ 1,763	\$ 1,908	\$ 3,671

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Three months ended September 30, 2019:			
Transaction and clearing, net	\$ 929	\$ —	\$ 929
Data services	—	553	553
Listings	—	114	114
Other revenues	67	—	67
Total revenues	996	667	1,663
Transaction-based expenses	327	—	327
Total revenues, less transaction-based expenses	<u>\$ 669</u>	<u>\$ 667</u>	<u>\$ 1,336</u>

Timing of Revenue Recognition

Services transferred at a point in time	\$ 578	\$ —	\$ 578
Services transferred over time	91	667	758
Total revenues, less transaction-based expenses	<u>\$ 669</u>	<u>\$ 667</u>	<u>\$ 1,336</u>

	Trading and Clearing Segment	Data and Listings Segment	Total Consolidated
Three months ended September 30, 2018:			
Transaction and clearing, net	\$ 760	\$ —	\$ 760
Data services	—	530	530
Listings	—	112	112
Other revenues	61	—	61
Total revenues	821	642	1,463
Transaction-based expenses	263	—	263
Total revenues, less transaction-based expenses	<u>\$ 558</u>	<u>\$ 642</u>	<u>\$ 1,200</u>

Timing of Revenue Recognition

Services transferred at a point in time	\$ 477	\$ —	\$ 477
Services transferred over time	81	642	723
Total revenues, less transaction-based expenses	<u>\$ 558</u>	<u>\$ 642</u>	<u>\$ 1,200</u>

The Trading and Clearing segment revenues above include \$188 million and \$184 million for the nine months ended September 30, 2019 and 2018, respectively, and \$66 million and \$56 million for the three months ended September 30, 2019 and 2018, respectively, for services transferred over time related to risk management of open interest performance obligations. A majority of these performance obligations are performed over a short period of time of one month or less.

Beginning in the second quarter of 2019, we have reflected amounts owed under certain third-party revenue share arrangements as technology and communication operating expenses rather than as had been previously recorded net within transaction and clearing revenues. These are included within our Trading and Clearing segment.

6. Goodwill and Other Intangible Assets

The following is a summary of the activity in the goodwill balance for the nine months ended September 30, 2019 (in millions):

Goodwill balance at December 31, 2018	\$ 13,085
Acquisitions	235
Foreign currency translation	(12)
Other activity, net	2
Goodwill balance at September 30, 2019	<u>\$ 13,310</u>

The following is a summary of the activity in the other intangible assets balance for the nine months ended September 30, 2019 (in millions):

Other intangible assets balance at December 31, 2018	\$	10,462
Acquisitions		116
Foreign currency translation		(13)
Amortization of other intangible assets		(235)
Other intangible assets balance at September 30, 2019	\$	10,330

We completed our acquisition of Simplifile and several other acquisitions during the nine months ended September 30, 2019 (Note 3). Foreign currency translation adjustments result from a portion of our goodwill and other intangible assets being held at our U.K., EU and Canadian subsidiaries, whose functional currencies are not the U.S. dollar. The change in other activity, net, in the goodwill table above primarily relates to adjustments to the fair value of the net tangible assets relating to acquisitions, with a corresponding adjustment to goodwill. We did not recognize any impairment losses on goodwill or other intangible assets during the nine and three months ended September 30, 2019.

7. Deferred Revenue

Our contract liabilities, or deferred revenue, represent consideration received that is yet to be recognized as revenue. Total deferred revenue was \$332 million as of September 30, 2019, including \$247 million in current deferred revenue and \$85 million in non-current deferred revenue. The changes in our deferred revenue during the nine months ended September 30, 2019 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Total
Deferred revenue balance at December 31, 2018	\$ —	\$ 25	\$ 100	\$ 92	\$ 217
Additions	387	11	30	300	728
Amortization	(289)	(17)	(30)	(277)	(613)
Deferred revenue balance at September 30, 2019	\$ 98	\$ 19	\$ 100	\$ 115	\$ 332

The changes in our deferred revenue during the nine months ended September 30, 2018 are as follows (in millions):

	Annual Listings Revenues	Original Listings Revenues	Other Listings Revenues	Data Services and Other Revenues	Total
Deferred revenue balance at December 31, 2017	\$ —	\$ 25	\$ 98	\$ 93	\$ 216
Additions	383	17	36	291	727
Amortization	(288)	(17)	(27)	(272)	(604)
Deferred revenue balance at September 30, 2018	\$ 95	\$ 25	\$ 107	\$ 112	\$ 339

Included in the amortization recognized during the nine months ended September 30, 2019 is \$94 million related to the deferred revenue balance as of January 1, 2019. Included in the amortization recognized for the nine months ended September 30, 2018 is \$98 million related to the deferred revenue balance as of January 1, 2018. As of September 30, 2019, the remaining deferred revenue balance for original listings revenue, other listings revenue and data services and other revenues will be recognized over the period of time we satisfy our performance obligations as described in Note 5.

8. Debt

Our total debt, including short-term and long-term debt, consisted of the following as of September 30, 2019 and December 31, 2018 (in millions):

	As of September 30, 2019	As of December 31, 2018
Debt:		
Short-term debt:		
Commercial Paper	\$ 1,318	\$ 951
Other short-term debt	11	—
Total short-term debt	1,329	951
Long-term debt:		
2020 Senior Notes (2.75% senior unsecured notes due December 1, 2020)	1,247	1,246
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	497	496
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	397	397
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	794	793
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,244	1,243
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	496	496
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	592	591
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,229	1,228
Total long-term debt	6,496	6,490
Total debt	\$ 7,825	\$ 7,441

Credit Facilities

We have a \$3.4 billion senior unsecured revolving credit facility, or the Credit Facility, with a maturity date of August 9, 2023. The Credit Facility includes an option for us to propose an increase in the aggregate amount available for borrowing by up to \$975 million, subject to the consent of the lenders funding the increase and certain other conditions. No amounts were outstanding under the Credit Facility as of September 30, 2019.

As of September 30, 2019, of the \$3.4 billion that is currently available for borrowing under the Credit Facility, \$1.3 billion is required to back-stop the amount outstanding under our U.S. dollar commercial paper program, or the Commercial Paper Program, and \$160 million is required to support certain broker-dealer subsidiary commitments. The amount required to back-stop the amounts outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$1.9 billion is available for working capital and general corporate purposes including, but not limited to, acting as a back-stop to future increases in the amounts outstanding under the Commercial Paper Program.

During the nine months ended September 30, 2019, a subsidiary of ours entered into a new \$20 million line of credit for their general corporate purposes. As of September 30, 2019, the subsidiary had borrowed \$11 million, which is reflected as “other short-term debt” in the table above.

Commercial Paper Program

Our Commercial Paper Program is currently backed by the borrowing capacity available under the Credit Facility, as described above. The effective interest rate of commercial paper issuances does not materially differ from short-term interest rates (such as USD LIBOR) which fluctuate due to market conditions and as a result may impact our interest expense. During the nine months ended September 30, 2019, we had net issuances of \$367 million under the Commercial Paper Program, the proceeds of which were used to fund the acquisition of Simplifile and for general corporate purposes.

Commercial paper notes of \$1.3 billion with original maturities ranging from one to 77 days were outstanding as of September 30, 2019 under our Commercial Paper Program, with a weighted average interest rate of 2.2% per annum and a weighted average maturity of 27 days.

9. Share-Based Compensation

We currently sponsor employee and director stock option, restricted stock and employee stock purchase plans. Stock options and restricted stock are granted at the discretion of the Compensation Committee of our Board of Directors based on the estimated fair value on the date of grant. The fair value of the stock options and restricted stock on the date of grant is recognized as expense over the vesting period, net of forfeitures. The non-cash compensation expenses recognized in our consolidated statements of income for stock options, restricted stock and under our employee stock purchase plan were \$100 million and \$93 million for the nine months ended September 30, 2019 and 2018, respectively, and \$36 million and \$32 million for the three months ended September 30, 2019 and 2018, respectively.

Stock Option Plans

The following is a summary of stock option activity for the nine months ended September 30, 2019:

	Number of Options (in thousands)	Weighted Average Exercise Price per Option
Outstanding at December 31, 2018	3,610	\$ 46.44
Granted	493	76.16
Exercised	(575)	38.71
Outstanding at September 30, 2019	3,528	51.86

Details of stock options outstanding as of September 30, 2019 are as follows:

	Number of Options (in thousands)	Weighted Average Exercise Price	Weighted Average Remaining Contractual Life (Years)	Aggregate Intrinsic Value (In millions)
Vested or expected to vest	3,528	\$ 51.86	6.4	\$ 143
Exercisable	2,458	\$ 44.27	5.4	\$ 118

The total intrinsic value of stock options exercised was \$25 million for both the nine months ended September 30, 2019 and 2018, and \$7 million and \$6 million for the three months ended September 30, 2019 and 2018, respectively. As of September 30, 2019, there were \$10 million in total unrecognized compensation costs related to stock options. These costs are expected to be recognized over a weighted average period of 1.6 years as the stock options vest.

We use the Black-Scholes option pricing model to value our stock option awards. During the nine months ended September 30, 2019 and 2018, we used the weighted-average assumptions in the table below to compute the value of all options for shares of common stock granted to employees:

Assumptions:	Nine Months Ended September 30,	
	2019	2018
Risk-free interest rate	2.49 %	2.67 %
Expected life in years	5.9	6.0
Expected volatility	20 %	20 %
Expected dividend yield	1.44 %	1.43 %
Estimated weighted-average fair value of options granted per share	\$ 15.45	\$ 14.08

The risk-free interest rate is based on the zero-coupon U.S. Treasury yield curve in effect at the date of grant. The expected life is derived from historical and anticipated future exercise patterns. Expected volatility is based on historical volatility data of our stock.

Restricted Stock Plans

Our restricted shares have vesting conditions based on company performance linked to both short-term and long-term stockholder return as well as retention objectives. The grant date fair value of our restricted stock awards is based on the closing stock price on the date of grant.

In February 2019, we reserved a maximum of 1.1 million restricted shares for potential issuance as performance-based restricted shares to certain of our employees. The number of shares that will ultimately be granted under this award will be based on our actual financial performance as compared to financial performance targets set by our Board of Directors and

the Compensation Committee of our Board of Directors for the year ending December 31, 2019, as well as our 2019 total stockholder return, or TSR, as compared to that of the S&P 500 Index. The maximum compensation expense to be recognized under these performance-based restricted shares is \$82 million if the maximum financial performance target is met and all 1.1 million shares vest. The compensation expense to be recognized under these performance-based restricted shares will be \$41 million if the target financial performance is met, which would result in 0.6 million shares vesting. We recognize expense on an accelerated basis over the three-year vesting period based on our quarterly assessment of the probable 2019 actual financial performance as compared to the 2019 financial performance targets. As of September 30, 2019, we determined that it is probable that the financial performance level will be at target for 2019. Based on this assessment, we recorded non-cash compensation expense of \$15 million and \$6 million for the nine and three months ended September 30, 2019, respectively, related to these shares and the remaining \$26 million in non-cash compensation expense will be recorded on an accelerated basis over the remaining vesting period, including \$7 million which will be recorded over the remainder of 2019.

The following is a summary of the non-vested restricted share activity for the nine months ended September 30, 2019:

	Number of Restricted Shares (in thousands)	Weighted Average Grant-Date Fair Value per Share
Non-vested at December 31, 2018	4,470	\$ 60.56
Granted	1,675	76.65
Vested	(2,191)	57.64
Forfeited	(189)	66.38
Non-vested at September 30, 2019	3,765	69.13

The shares granted include 1 million time-based restricted shares and the remainder are performance-based. Performance-based restricted shares have been presented to reflect the actual shares to be issued based on the achievement of past performance targets, also considering the impact of any market conditions. Non-vested performance-based restricted shares granted are presented in the table above at the target number of restricted shares that would vest if the performance targets are met. As of September 30, 2019, there were \$137 million in total unrecognized compensation costs related to time-based and performance-based restricted stock. These costs are expected to be recognized over a weighted-average period of 1.4 years as the restricted stock vests. These unrecognized compensation costs assume that a target performance level will be met on the performance-based restricted shares granted in February 2019. During the nine months ended September 30, 2019 and 2018, the total fair value of restricted stock vested under all restricted stock plans was \$166 million and \$201 million, respectively.

Bakkt Incentive Units

In February 2019, our Board approved the adoption of the Bakkt Equity Incentive Plan to issue various Bakkt equity unit awards. Under this plan, during the nine months ended September 30, 2019, Bakkt issued 101 million, 4 million and 8 million of its preferred, common and phantom incentive units, respectively, to certain employees and Board members. The issued units are unvested at the issuance date, are subject to the vesting terms in the award agreements and upon vesting are converted into Bakkt equity or cash. With the assistance of third-party valuation experts and based on our assumptions as of the issuance date, we estimate that approximately \$46 million of compensation expense will be recognized over an eight-year period associated with these awards.

10. Equity

Stock Repurchase Program

In September 2018, our Board of Directors approved an aggregate of \$2.0 billion for repurchases of our common stock with no fixed expiration date that became effective January 1, 2019. During the nine months ended September 30, 2019, we repurchased 12.5 million shares of our outstanding common stock at a cost of \$1.0 billion under our Rule 10b5-1 trading plan and 1.3 million shares at a cost of \$100 million on the open market. As of September 30, 2019, up to \$880 million remains from the Board authorization for repurchases of our common stock. We expect to fund repurchases from our operating cash flow, borrowings under our debt facilities or our Commercial Paper Program. Repurchases may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. We have entered into a Rule 10b5-1 trading plan, as authorized by our Board, to govern some of the repurchases of our shares of common stock. We may discontinue the stock repurchases at any time and may amend or terminate the Rule 10b5-1 trading plan at any time. The approval of our Board for the share

repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board of Directors may increase or decrease the amount available for repurchases from time to time.

Dividends

During the nine months ended September 30, 2019 and 2018, we declared and paid cash dividends per share of \$0.825 and \$0.72, respectively, for an aggregate payout of \$467 million and \$417 million, respectively, and during the three months ended September 30, 2019 and 2018, we declared and paid cash dividends per share of \$0.275 and \$0.24, respectively, for an aggregate payout of \$155 million and \$138 million, respectively. The declaration of dividends is subject to the discretion of our Board, and may be affected by various factors, including our future earnings, financial condition, capital requirements, levels of indebtedness, credit ratings and other considerations which our Board of Directors deem relevant. Our Board has adopted a quarterly dividend declaration policy providing that the declaration of any dividends will be determined quarterly by the Board or its Audit Committee, taking into account such factors as our evolving business model, prevailing business conditions and our financial results and capital requirements, without a predetermined annual net income payout ratio.

Accumulated Other Comprehensive Income (Loss)

The following tables present changes in the accumulated balances for each component of other comprehensive income (loss) (in millions):

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2018	\$ (227)	\$ 2	\$ (90)	\$ (315)
Other comprehensive income (loss)	(33)	(1)	—	(34)
Income tax benefit (expense)	1	—	—	1
Net current period other comprehensive income (loss)	(32)	(1)	—	(33)
Balance, as of September 30, 2019	\$ (259)	\$ 1	\$ (90)	\$ (348)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of June 30, 2019	\$ (220)	\$ 1	\$ (90)	\$ (309)
Other comprehensive income (loss)	(40)	—	—	(40)
Income tax benefit (expense)	1	—	—	1
Net current period other comprehensive income (loss)	(39)	—	—	(39)
Balance, as of September 30, 2019	\$ (259)	\$ 1	\$ (90)	\$ (348)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of December 31, 2017	\$ (136)	\$ 2	\$ (89)	\$ (223)
Other comprehensive income (loss)	(52)	—	—	(52)
Income tax benefit (expense)	1	—	—	1
Net current period other comprehensive income (loss)	(51)	—	—	(51)
Balance, as of September 30, 2018	\$ (187)	\$ 2	\$ (89)	\$ (274)

Changes in Accumulated Other Comprehensive Income (Loss) by Component				
	Foreign currency translation adjustments	Comprehensive income from equity method investment	Employee benefit plans adjustments	Total
Balance, as of June 30, 2018	\$ (178)	\$ 2	\$ (89)	\$ (265)
Other comprehensive income (loss)	(9)	—	—	(9)
Income tax benefit (expense)	—	—	—	—
Net current period other comprehensive income (loss)	(9)	—	—	(9)
Balance, as of September 30, 2018	\$ (187)	\$ 2	\$ (89)	\$ (274)

11. Income Taxes

Our effective tax rate was 20% and 21% for the nine months ended September 30, 2019 and 2018, respectively, and 16% for both the three months ended September 30, 2019 and 2018. The effective tax rate for the nine months ended September 30,

2019 is lower than the effective tax rate for the same period in 2018 primarily due to changes in estimates of certain international tax provisions enacted as part of the U.S. Federal Tax Cuts and Jobs Act, or TCJA, and reductions of income tax reserves resulting from the expiration of certain statutes of limitation.

12. Clearing Operations

We operate six clearing houses, each of which acts as a central counterparty that becomes the buyer to every seller and the seller to every buyer for its clearing members. Through this central counterparty function, the clearing houses provide financial security for each transaction for the duration of the position by limiting counterparty credit risk.

Our clearing houses are responsible for providing clearing services to each of our futures exchanges, and in some cases outside of our execution venues, and are as follows, referred to herein collectively as "the ICE Clearing Houses":

Clearing House	Products Cleared	Exchange where Executed	Location
ICE Clear Europe	Energy, agricultural, interest rates and equity index futures and options contracts and OTC European CDS instruments	ICE Futures Europe, ICE Futures U.S., ICE Endex and third-party venues	U.K.
ICE Clear U.S.	Agricultural, metals, FX and equity index futures and options contracts	ICE Futures U.S.	U.S.
ICE Clear Credit	North American, European, Asian-Pacific and Emerging Market CDS instruments	Creditex, OTC and third-party venues	U.S.
ICE Clear Netherlands	Derivatives on equities and equity indices traded on regulated markets	ICE Endex	The Netherlands
ICE Clear Singapore	Energy, metals and financial futures products	ICE Futures Singapore	Singapore
ICE NGX	Physical North American natural gas, electricity and oil futures	ICE NGX	Canada

Original & Variation Margin

Each of the ICE Clearing Houses generally requires all clearing members or participants to deposit collateral in cash or certain pledged assets. The collateral deposits are known as "original margin." In addition, the ICE Clearing Houses may make intraday original margin calls in circumstances where market conditions require additional protection. The daily profits and losses to and from the ICE Clearing Houses due to the marking-to-market of open contracts is known as "variation margin." With the exception of ICE NGX's physical natural gas and physical power products discussed separately below, the ICE Clearing Houses mark all outstanding contracts to market, and therefore pay and collect variation margin, at least once daily.

The amounts that the clearing members and participants are required to maintain are determined by proprietary risk models established by each ICE Clearing House and reviewed by the relevant regulators, independent model validators, risk committees and the boards of directors of the respective ICE Clearing House. The amounts required may fluctuate over time. Each of the ICE Clearing Houses is a separate legal entity and is not subject to the liabilities of the others, or the obligations of the members of the other ICE Clearing Houses.

Should a particular clearing member or participant fail to deposit its original margin or fail to make a variation margin payment, when and as required, the relevant ICE Clearing House may liquidate or hedge its open positions and use their original margin and guaranty fund deposits to pay any amount owed. In the event that the defaulting clearing member's deposits are not sufficient to pay the amount owed in full, the ICE Clearing Houses will first use their respective contributions to the guaranty fund, often referred to as Skin In The Game, or SITG, to pay any remaining amount owed. In the event that the SITG is not sufficient, the ICE Clearing Houses may utilize the respective guaranty fund deposits, or collect additional funds from their respective non-defaulting clearing members on a pro-rata basis, to pay any remaining amount owed.

As of September 30, 2019 and December 31, 2018, the ICE Clearing Houses have received or have been pledged \$122.1 billion and \$121.4 billion, respectively, in cash and non-cash collateral in original margin and guaranty fund deposits to cover price movements of underlying contracts for both periods.

Guaranty Funds & ICE Contribution

As described above, mechanisms have been created, called guaranty funds, to provide partial protection in the event of a clearing member default. With the exception of ICE NGX, each of the ICE Clearing Houses requires that each clearing member make deposits into a guaranty fund.

In addition, we have contributed our own capital which could be used if a defaulting clearing member's original margin and guaranty fund deposits are insufficient. Such amounts are recorded as long-term restricted cash and cash equivalents in our balance sheets and are as follows (in millions):

Clearing House	ICE Portion of Guaranty Fund Contribution		Default insurance	
	As of September 30, 2019	As of December 31, 2018	As of September 30, 2019	As of December 31, 2018
ICE Clear Europe	\$233	\$206	\$75	N/A
ICE Clear U.S.	103	61	25	N/A
ICE Clear Credit	50	50	50	N/A
ICE Clear Netherlands	2	2	N/A	N/A
ICE Clear Singapore	1	1	N/A	N/A
ICE NGX	N/A	N/A	100	\$100
Total	\$389	\$320	\$250	\$100

In January 2019, we increased our contribution to ICE Clear Europe's guaranty fund by \$27 million, in March 2019, we increased our ICE Clear U.S. guaranty fund contribution by \$7 million and in September 2019, in connection with the launch of Bakkt, Bakkt contributed \$35 million to the ICE Clear U.S. guaranty fund, solely applicable to any losses associated with a default in Bitcoin contracts and other digital asset contracts that ICE Clear U.S. might clear in the future.

In September 2019, we added a layer of insurance to our clearing member default protection. The default insurance layer resides after and in addition to the ICE Clear Credit, ICE Clear Europe, and ICE Clear U.S. SITG contributions and before the guaranty fund contributions of the non-defaulting clearing members. The default insurance has a three-year term, subject to renewal.

Similar to SITG, the default insurance layer is not intended to replace or reduce the position risk-based amount of the guaranty fund. As a result, the default insurance layer is not a factor that is included in the calculation of the clearing members' guaranty fund contribution requirement. Instead, it serves as a new, additional, distinct, and separate default resource that should serve to further protect the non-defaulting clearing members' guaranty fund contributions from being mutualized in the event of a default.

As of September 30, 2019, ICE NGX maintains a guaranty fund utilizing a \$100 million letter of credit and a default insurance policy, discussed below.

Cash and Cash Equivalent Deposits

We have recorded cash and cash equivalent margin deposits and amounts due in our balance sheets as current assets with corresponding current liabilities to the clearing members. As of September 30, 2019, our cash and cash equivalent margin deposits are as follows (in millions):

	ICE Clear Europe ⁽¹⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 28,971	\$ 22,534	\$ 7,157	\$ —	\$ 5	\$ 58,667
Unsettled variation margin, net	—	—	—	229	—	229
Guaranty fund	4,198	2,120	445	—	5	6,768
Delivery contracts receivable/payable, net	—	—	—	324	—	324
Total	\$ 33,169	\$ 24,654	\$ 7,602	\$ 553	\$ 10	\$ 65,988

As of December 31, 2018, our cash and cash equivalent deposits, are as follows (in millions):

	ICE Clear Europe ⁽²⁾	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Other ICE Clearing Houses	Total
Original margin	\$ 27,597	\$ 22,770	\$ 6,260	\$ —	\$ 3	\$ 56,630
Unsettled variation margin, net	—	—	—	417	—	417
Guaranty fund	3,267	2,456	460	—	5	6,188
Delivery contracts receivable/payable, net	—	—	—	720	—	720
Total	\$ 30,864	\$ 25,226	\$ 6,720	\$ 1,137	\$ 8	\$ 63,955

⁽¹⁾ \$27.9 billion and \$5.3 billion is related to futures/options and CDS, respectively.

⁽²⁾ \$25.8 billion and \$5.1 billion is related to futures/options and CDS, respectively.

Our cash and cash equivalent margin and guaranty fund deposits are maintained in accounts with national banks and reputable financial institutions or secured through direct investments, primarily in U.S. Treasury securities with original maturities of less than three months, or reverse repurchase agreements with primarily overnight maturities.

To provide a tool to address the liquidity needs of our clearing houses and manage the liquidation of margin and guaranty fund deposits held in the form of cash and high quality sovereign debt, ICE Clear Europe, ICE Clear Credit and ICE Clear U.S. have entered into Committed Repurchase Agreement Facilities, or Committed Repo. Additionally, ICE Clear Credit and ICE Clear Netherlands have entered into Committed FX Facilities to support these liquidity needs. As of September 30, 2019 the following facilities were in place:

- **ICE Clear Europe:** \$1.0 billion in Committed Repo to finance U.S. dollar, euro and pound sterling deposits.
- **ICE Clear Credit:** \$300 million in Committed Repo to finance U.S. dollar and euro deposits, €250 million in Committed Repo to finance euro deposits, and €1.9 billion in Committed FX Facilities to finance euro payment obligations.
- **ICE Clear U.S.:** \$250 million in Committed Repo to finance U.S. dollar deposits.
- **ICE Clear Netherlands:** €10 million in Committed FX Facilities to finance euro payment obligations.

Details of our cash and cash equivalent deposits are as follows (in millions):

Clearing House	Investment Type	As of September 30, 2019	As of December 31, 2018
ICE Clear Europe	National Bank Account ⁽¹⁾	\$ 5,886	\$ 8,647
ICE Clear Europe	Reverse repo	24,184	18,097
ICE Clear Europe	Sovereign Debt	2,940	4,035
ICE Clear Europe	Demand deposits	159	85
ICE Clear Credit	National Bank Account ⁽²⁾	19,947	19,484
ICE Clear Credit	Reverse repo	2,997	1,935
ICE Clear Credit	Demand deposits	1,710	3,807
ICE Clear U.S.	Reverse repo	5,061	4,380
ICE Clear U.S.	U.S. Treasuries	2,541	2,340
Other ICE Clearing Houses	Demand deposits	10	8
ICE NGX	Unsettled Variation Margin and Delivery Contracts Receivable/Payable	553	1,137
Total		\$ 65,988	\$ 63,955

⁽¹⁾ As of September 30, 2019, ICE Clear Europe held €4.1 billion (\$4.4 billion based on the euro/U.S. dollar exchange rate of 1.0899 as of September 30, 2019) at De Nederlandsche Bank, or DNB, £1.1 million (\$1.4 million based on the pound sterling/U.S. dollar exchange rate of 1.2295 as of September 30, 2019) at the Bank of England, or BOE and €10 million (\$11 million based on the above exchange rate) at the BOE. As of December 31, 2018, ICE Clear Europe held €7.0 billion (\$8.0 billion based on the euro/U.S. dollar exchange rate of 1.1466 as of December 31, 2018) at DNB and £500 million (\$638 million based on the pound sterling/U.S. dollar exchange rate of 1.2756 as of December 31, 2018) at the BOE.

⁽²⁾ ICE Clear Credit is a systemically important financial market utility, or SIFMU, as designated by the Financial Stability Oversight Council, and holds its U.S. dollar cash margin in cash accounts at the Federal Reserve Bank of Chicago.

Other Deposits

In addition to the cash deposits above, the ICE Clearing Houses have also received other assets from clearing members, which include government obligations, and may include other cash deposits and non-cash collateral such as letters of credit or gold to mitigate credit risk. For certain deposits, we may impose discount or “haircut” rates to ensure adequate collateral if market values fluctuate. The value-related risks and rewards of these assets remain with the clearing members. Any gain or loss accrues to the clearing member. The ICE Clearing Houses do not rehypothecate or repledge these assets. These pledged assets are not reflected in our balance sheets, and are as follows (in millions):

As of September 30, 2019					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 29,305	\$ 13,420	\$ 9,881	\$ —	\$ 52,606
Letters of credit	—	—	—	2,247	2,247
ICE NGX cash deposits	—	—	—	322	322
Total	\$ 29,305	\$ 13,420	\$ 9,881	\$ 2,569	\$ 55,175
Guaranty fund:					
Government securities at face value	\$ 407	\$ 284	\$ 274	\$ —	\$ 965
As of December 31, 2018					
	ICE Clear Europe	ICE Clear Credit	ICE Clear U.S.	ICE NGX	Total
Original margin:					
Government securities at face value	\$ 29,887	\$ 12,990	\$ 10,208	\$ —	\$ 53,085
Letters of credit	—	—	—	2,556	2,556
ICE NGX cash deposits	—	—	—	605	605
Total	\$ 29,887	\$ 12,990	\$ 10,208	\$ 3,161	\$ 56,246
Guaranty fund:					
Government securities at face value	\$ 654	\$ 256	\$ 264	\$ —	\$ 1,174

ICE NGX

ICE NGX is the central counterparty to participants on opposite sides of its physically-settled contracts, and the balance related to delivered but unpaid contracts is recorded as a delivery contract net receivable, with an offsetting delivery contract net payable in our balance sheets. Unsettled variation margin equal to the fair value of open contracts is recorded as of each balance sheet date. ICE NGX marks all outstanding contracts to market daily, but only collects variation margin when a clearing member's or participant's open position falls outside a specified percentage of its pledged collateral.

ICE NGX requires participants to maintain cash or letters of credit to serve as collateral in the event of default. The cash is maintained in a segregated bank account, held in trust and remains the property of the participant; therefore, it is not included in our balance sheets. ICE NGX maintains the following accounts with a third-party Canadian chartered bank which are available in the event of physical settlement shortfalls, subject to certain conditions:

Account Type	As of September 30, 2019 (In C\$ millions)	As of September 30, 2019 (In \$USD millions)
Daylight liquidity facility	C\$300	\$227
Overdraft facility	20	15
Total	C\$320	\$242

As of September 30, 2019, ICE NGX maintains a guaranty fund utilizing a \$100 million letter of credit that has been entered into with a major Canadian chartered bank and backed by a default insurance policy underwritten by Export Development Corporation, or EDC, a Canadian government agency. In the event of a participant default, where a participant's collateral becomes depleted, the remaining shortfall would be covered by a draw down on the letter of credit following which ICE NGX would pay the first \$15 million in losses per its deductible and recover additional losses under the insurance policy up to \$100 million.

Clearing House Exposure

Each ICE Clearing House bears financial counterparty credit risk and provides a central counterparty guarantee, or performance guarantee, to its clearing members or participants. To reduce their exposure, the ICE Clearing Houses have a risk management program with both initial and ongoing membership standards. Excluding the effects of original and variation margin, guaranty fund and collateral requirements, the ICE Clearing Houses' maximum estimated exposure for this guarantee is \$105.6 billion as of September 30, 2019, which represents the maximum estimated value by the ICE Clearing Houses of a hypothetical one-day movement in pricing of the underlying unsettled contracts. This value was determined using proprietary risk

management software that simulates gains and losses based on historical market prices, volatility and other factors present at that point in time for those particular unsettled contracts. Future actual market price volatility could result in the exposure being significantly different than this amount.

13. Legal Proceedings

We are subject to legal proceedings, claims and investigations that arise in the ordinary course of our business. These include the matters described in Part I, Item 3 “Legal Proceedings” and Note 15 to the consolidated financial statements in Part II, Item 8 of our 2018 Form 10-K. We establish accruals for those matters in circumstances when a loss contingency is considered probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. We do not believe that the resolution of these legal matters, including the matters described below, will have a material adverse effect on our consolidated financial condition, results of operations, or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period could be materially and adversely affected by any developments relating to the legal proceedings, claims and investigations.

LIBOR Litigation

On January 15, 2019 and January 31, 2019, two virtually identical purported class action complaints were filed by, respectively, Putnam Bank, a savings bank based in Putnam, Connecticut, and two municipal pension funds affiliated with the City of Livonia, Michigan in the U.S. District Court for the Southern District of New York against ICE and several of its subsidiaries, including ICE Benchmark Administration Limited (“IBA”) (the “ICE Defendants”), as well as 18 multinational banks and various of their respective subsidiaries and affiliates (the “Panel Bank Defendants”). On March 4, 2019, a virtually identical complaint was filed on behalf of four retirement and benefit funds affiliated with the Hawaii Sheet Metal Workers Union. IBA is the administrator for various regulated benchmarks, including the ICE LIBOR benchmark that is calculated daily based upon the submissions from a reference panel (which includes the Panel Bank Defendants). On July 1, 2019, the various plaintiffs referenced above filed a consolidated amended complaint against the ICE and Panel Bank Defendants.

The plaintiffs seek to litigate on behalf of a purported class of all U.S.-based persons or entities who transacted with a Panel Bank Defendant by receiving a payment on an interest rate indexed to a one-month or three-month USD LIBOR-benchmarked rate during the period February 1, 2014 to the present. The plaintiffs allege that the ICE and Panel Bank Defendants engaged in a conspiracy to set the LIBOR benchmark at artificially low levels, with an alleged purpose and effect of depressing payments by the Panel Bank Defendants to members of the purported class.

As with the individual complaints, the consolidated amended complaint asserts a claim for violations of the Sherman and Clayton Antitrust Acts and seeks unspecified treble damages and other relief. The ICE and Panel Bank Defendants filed motions to dismiss the consolidated amended complaint on August 30, 2019 and briefing of these motions is not yet complete. ICE intends to vigorously defend the matter.

City of Providence Litigation

In the City of Providence litigation, the district court entered an order on May 28, 2019 denying the motions to dismiss filed by the defendant exchanges (which include New York Stock Exchange LLC and NYSE Arca, Inc., two of our subsidiaries). The exchanges filed a motion in the district court on June 17, 2019 asking the court to certify the matter for an immediate appeal to the U.S. Court of Appeals for the Second Circuit; on July 16, 2019 the court denied the motion. On July 25, 2019, the exchanges filed answers to the second amended complaint, denying the principal allegations of the plaintiffs, denying liability in the matter, and asserting various affirmative defenses. The discovery period in the matter has commenced and is scheduled to continue into 2020. For further information on the City of Providence litigation, see our 2018 Form 10-K.

14. Fair Value Measurements

Fair value is the price that would be received from selling an asset or paid to transfer a liability (an exit price) in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants at the measurement date. Our financial instruments consist primarily of certain short-term and long-term assets and liabilities, customer accounts receivable, margin deposits and guaranty funds, equity investments, and short-term and long-term debt.

The fair value of our financial instruments is measured based on a three-level hierarchy:

- **Level 1 inputs** — quoted prices for identical assets or liabilities in active markets.
- **Level 2 inputs** — observable inputs other than Level 1 inputs such as quoted prices for similar assets and liabilities in active markets or inputs other than quoted prices that are directly observable.

- **Level 3 inputs** — unobservable inputs supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

Financial assets and liabilities recorded or disclosed at fair value in the accompanying consolidated balance sheets as of September 30, 2019 and December 31, 2018 are classified in their entirety based on the lowest level of input that is significant to the asset or liability's fair value measurement.

Our mutual funds are equity and fixed income mutual funds held for the purpose of providing future payments for the supplemental executive savings plan and the supplemental executive retirement plan and are classified as equity investments and measured at fair value using Level 1 inputs with adjustments recorded in net income.

We hold money market funds measured at fair value using Level 1 inputs with adjustments recorded in net income.

MERS is part of our ICE Mortgage Services business and holds fixed income investments as part of a reserve fund in order to satisfy the original terms of the governing documents of our June 2016 acquisition of a majority equity position in MERS. These investments are measured at fair value using Level 2 inputs with adjustments recorded in net income. In June 2019 we sold \$41 million of these investments in anticipation of the payment due to the original shareholders of MERS in accordance with the acquisition agreement. The proceeds from the sale of the investments were distributed to the original shareholders of MERS in July 2019.

Excluding our equity investments without a readily determinable fair value, all other financial instruments are determined to approximate carrying value due to the short period of time to their maturities.

We did not use Level 3 inputs to determine the fair value of assets or liabilities measured at fair value on a recurring basis as of September 30, 2019 or December 31, 2018.

We measure certain assets, such as intangible assets, at fair value on a non-recurring basis. These assets are recognized at fair value if they are deemed to be impaired. As of September 30, 2019 and December 31, 2018, none of our intangible assets were required to be recorded at fair value since no impairments were recorded.

We measure certain equity investments at fair value on a non-recurring basis using our policy election under ASU No. 2016-01, *Financial Instruments-Overall (Subtopic 825-10): Recognition and Measurement of Financial Assets and Financial Liabilities*. During the nine months ended September 30, 2019, we evaluated a transaction involving one of these investments and concluded that no fair value adjustment was required under this election.

See Note 12 for the fair value considerations related to our margin deposits, guaranty funds and delivery contracts receivable.

The table below displays the fair value of our debt as of September 30, 2019. The fair values of our fixed rate notes were estimated using quoted market prices for these instruments. The fair value of our commercial paper includes a discount and other short-term debt approximates par value since the interest rates on this short-term debt approximate market rates as of September 30, 2019.

Debt:	As of September 30, 2019	
	(in millions)	
	Carrying Amount	Fair value
Commercial Paper	\$ 1,318	\$ 1,323
Other short-term debt	11	11
2020 Senior Notes (2.75% senior unsecured notes due December 1, 2020)	1,247	1,260
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	497	505
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	397	419
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	794	855
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,244	1,349
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	496	524
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	592	660
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,229	1,493
Total debt	\$ 7,825	\$ 8,399

15. Segment Reporting

We operate two business segments: our Trading and Clearing segment and our Data and Listings segment. This presentation is reflective of how our chief operating decision maker reviews and operates our business. Our Trading and Clearing segment comprises our transaction-based execution and clearing businesses. Our Data and Listings segment comprises our data services and our securities listings businesses, which are both largely subscription-based. Our chief operating decision maker does not review total assets or statements of income below operating income by segments; therefore, such information is not presented below. Our two segments do not engage in intersegment transactions.

Financial data for our business segments is as follows for the nine and three months ended September 30, 2019 and 2018 (in millions):

	Nine Months Ended September 30, 2019			Nine Months Ended September 30, 2018		
	Trading and Clearing Segment	Data and Listings Segment	Consolidated	Trading and Clearing Segment	Data and Listings Segment	Consolidated
Revenues:						
Energy futures and options contracts	\$ 749	\$ —	\$ 749	\$ 708	\$ —	\$ 708
Agricultural and metals futures and options contracts	194	—	194	197	—	197
Financial futures and options contracts	252	—	252	262	—	262
Cash equities and equity options	1,201	—	1,201	1,162	—	1,162
Fixed income and credit	268	—	268	157	—	157
OTC and other transactions	34	—	34	36	—	36
Pricing and analytics	—	809	809	—	779	779
Exchange data and feeds	—	528	528	—	496	496
Desktops and connectivity	—	315	315	—	301	301
Listings	—	336	336	—	332	332
Other revenues	194	—	194	169	—	169
Revenues	2,892	1,988	4,880	2,691	1,908	4,599
Transaction-based expenses	976	—	976	928	—	928
Revenues, less transaction-based expenses	1,916	1,988	3,904	1,763	1,908	3,671
Operating expenses	732	1,121	1,853	655	1,109	1,764
Operating income	\$ 1,184	\$ 867	\$ 2,051	\$ 1,108	\$ 799	\$ 1,907

	Three Months Ended September 30, 2019			Three Months Ended September 30, 2018		
	Trading and Clearing Segment	Data and Listings Segment	Consolidated	Trading and Clearing Segment	Data and Listings Segment	Consolidated
Revenues:						
Energy futures and options contracts	\$ 265	\$ —	\$ 265	\$ 223	\$ —	\$ 223
Agricultural and metals futures and options contracts	60	—	60	58	—	58
Financial futures and options contracts	91	—	91	77	—	77
Cash equities and equity options	401	—	401	335	—	335
Fixed income and credit	101	—	101	56	—	56
OTC and other transactions	11	—	11	11	—	11
Pricing and analytics	—	273	273	—	263	263
Exchange data and feeds	—	172	172	—	168	168
Desktops and connectivity	—	108	108	—	99	99
Listings	—	114	114	—	112	112
Other revenues	67	—	67	61	—	61
Revenues	996	667	1,663	821	642	1,463
Transaction-based expenses	327	—	327	263	—	263
Revenues, less transaction-based expenses	669	667	1,336	558	642	1,200
Operating expenses	255	375	630	230	368	598
Operating income	\$ 414	\$ 292	\$ 706	\$ 328	\$ 274	\$ 602

Revenue from one clearing member of the Trading and Clearing segment comprised \$286 million, or 15% and \$92 million, or 14% of our Trading and Clearing revenues, less transaction-based expenses for the nine and three months ended September 30, 2019, respectively. Revenue from one clearing member of the Trading and Clearing segment comprised \$300 million, or 17%, and \$89 million, or 16% of our Trading and Clearing revenues, less transaction-based expenses for the nine and three months ended September 30, 2018, respectively. Clearing members are primarily intermediaries and represent a broad range of principal trading firms. If a clearing member ceased its operations, we believe that the trading firms would continue to conduct transactions and would clear those transactions through another clearing member firm. No additional customers or clearing members accounted for more than 10% of our segment revenues or consolidated revenues for the nine and three months ended September 30, 2019 and 2018.

16. Earnings Per Common Share

The following is a reconciliation of the numerators and denominators of the basic and diluted earnings per common share computations for the nine and three months ended September 30, 2019 and 2018 (in millions, except per share amounts):

	Nine Months Ended September 30,		Three Months Ended September 30,	
	2019	2018	2019	2018
Basic:				
Net income attributable to Intercontinental Exchange, Inc.	\$ 1,485	\$ 1,377	\$ 529	\$ 458
Weighted average common shares outstanding	563	577	559	572
Basic earnings per common share	\$ 2.64	\$ 2.39	\$ 0.95	\$ 0.80
Diluted:				
Weighted average common shares outstanding	563	577	559	572
Effect of dilutive securities - stock options and restricted shares	3	4	4	4
Diluted weighted average common shares outstanding	566	581	563	576
Diluted earnings per common share	\$ 2.62	\$ 2.37	\$ 0.94	\$ 0.79

Basic earnings per common share is calculated using the weighted average common shares outstanding during the period. Common equivalent shares from stock options and restricted stock awards, using the treasury stock method, are included in the diluted per share calculations unless the effect of their inclusion would be antidilutive. During the nine months ended

September 30, 2019 and 2018, 437,000 and 451,000 outstanding stock options, respectively, were not included in the computation of diluted earnings per common share, because to do so would have had an antidilutive effect. Certain figures in the table above may not recalculate due to rounding.

17. Subsequent Events

We have evaluated subsequent events and determined that no other events or transactions met the definition of a subsequent event for purposes of recognition or disclosure in the accompanying consolidated financial statements.

ITEM 2. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

In this Quarterly Report on Form 10-Q, unless otherwise indicated, the terms "Intercontinental Exchange," "ICE," "we," "us," "our," "our company" and "our business" refer to Intercontinental Exchange, Inc., together with its consolidated subsidiaries. References to "ICE Products" mean products listed on one or more of our markets. Due to rounding, figures may not sum exactly.

Forward-Looking Statements

This Quarterly Report on Form 10-Q, including the sections entitled "Notes to Consolidated Financial Statements," "Legal Proceedings" and "Management's Discussion and Analysis of Financial Condition and Results of Operations," contains "forward-looking statements" as defined in the Private Securities Litigation Reform Act of 1995. Any statements contained herein that are not statements of historical fact may be forward-looking statements. Any forward looking statements are based on our present beliefs and assumptions as well as the information currently available to us. Forward-looking statements may be introduced by or contain terminology such as "may," "will," "should," "could," "would," "targets," "goal," "expect," "intend," "plan," "anticipate," "believe," "estimate," "predict," "potential," "continue," or the antonyms of these terms or other comparable terminology. These forward-looking statements relate to future events or our future financial performance and involve known and unknown risks, uncertainties and other factors that may cause our results, levels of activity, performance, cash flows, financial position or achievements to differ materially from those expressed or implied by these statements. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, cash flows, financial position or achievements. Accordingly, we caution you not to place undue reliance on any forward-looking statements we may make.

Factors that may affect our performance and the accuracy of any forward-looking statements include, but are not limited to, those listed below:

- conditions in global financial markets and domestic and international economic, political and social conditions;
- the impact of the introduction of or any changes in laws, regulations, rules or government policies with respect to financial markets, increased regulatory scrutiny or enforcement actions and our ability to comply with these requirements;
- volatility in commodity prices, equity prices and price volatility of financial benchmarks and instruments such as interest rates, credit spreads, equity indices and foreign exchange rates;
- the business environment in which we operate and trends in our industry, including trading volumes, clearing, data services, fees, changing regulations, competition and consolidation;
- our ability to minimize the risks associated with operating clearing houses in multiple jurisdictions;
- our equity and options exchanges' compliance with their respective regulatory and oversight responsibilities;
- the resilience of our electronic platforms and soundness of our business continuity and disaster recovery plans;
- changes in renewal rates of subscription-based data revenues;
- our ability to execute our growth strategy, identify and effectively pursue, implement and integrate acquisitions and strategic alliances and realize the synergies and benefits of such transactions within the expected time frame;
- the performance and reliability of our trading and clearing technologies and those of third-party service providers;
- our ability to keep pace with technological developments and client preferences;
- our ability to ensure that the technology we utilize is not vulnerable to cyber attacks, hacking and other cybersecurity risks or other disruptive events;
- our ability to identify trends and adjust our business to benefit from such trends;
- our ability to evolve our benchmarks and indices in a manner that maintains or enhances their relevance;
- the accuracy of our cost and other financial estimates and our belief that cash flows from operations will be sufficient to service our debt and to fund our operational and capital expenditure needs;
- our ability to secure additional debt;
- our ability to maintain existing market participants and data customers, and to attract new ones;
- our ability to offer additional products and services, leverage our risk management capabilities and enhance our technology in a timely and cost-effective fashion;
- our ability to attract and retain key talent;
- our ability to protect our intellectual property rights and to operate our business without violating the intellectual property rights of others;
- potential adverse results of threatened or pending litigation and regulatory actions and proceedings; and
- our ability to realize the expected benefits of our majority investment in Bakkt which could result in additional unanticipated costs and risks.

These risks and other factors include those set forth in Item 1(A) under the caption “Risk Factors” in our Annual Report on Form 10-K for the year ended December 31, 2018, or our 2018 Form 10-K, as filed with the SEC on February 7, 2019. Due to the uncertain nature of these factors, management cannot assess the impact of each factor on the business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements.

Any forward-looking statement speaks only as of the date on which such statement is made, and we undertake no obligation to update any of these statements to reflect events or circumstances occurring after the date of this Quarterly Report on Form 10-Q. New factors may emerge and it is not possible to predict all factors that may affect our business and prospects.

Overview

We are a leading global operator of regulated exchanges, clearing houses and listings venues, and a provider of data services for commodity, financial, fixed income and equity markets. We operate regulated marketplaces for listing, trading and clearing of a broad array of derivatives contracts and securities across major asset classes, including energy and agricultural commodities, metals, interest rates, equities, ETFs, credit derivatives, digital assets, bonds and currencies, as well as offer mortgage and technology services. We also offer comprehensive data services to support the trading, investment, risk management and connectivity needs of customers around the world and across asset classes.

Our exchanges include derivative exchanges in the U.S., U.K., EU, Canada and Singapore, and cash equities, equity options and bond trading venues in the U.S. We also operate OTC markets for physical energy, fixed income and CDS trade execution. To serve global derivatives markets, we operate central counterparty clearing houses, or CCPs, in the U.S., U.K., EU, Canada and Singapore. We offer a range of data services, globally, for financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through our markets, clearing houses, listings venues and data services, we provide comprehensive solutions for our customers to manage risk and raise capital through liquid markets, benchmark products, access to capital markets and related services. Our business is conducted as two reportable business segments, our Trading and Clearing segment and our Data and Listings segment, and the majority of our identifiable assets are located in the U.S. and U.K.

Recent Developments

Launch of Bakkt

In August 2019, Bakkt received approval by the New York State Department of Financial Services to create Bakkt Trust Company, LLC, a qualified custodian. On September 23, 2019, the Bakkt Bitcoin Futures contract launched, with contracts traded on ICE Futures U.S. and cleared through ICE Clear U.S. Bakkt Trust Company, LLC takes custody of bitcoin for physically delivered futures, creating the first fully regulated marketplace to transact physically-delivered digital assets futures. In connection with the launch of Bakkt, in September 2019, Bakkt contributed \$35 million to our ICE Clear U.S. guaranty fund.

Acquisitions

On June 12, 2019, we acquired Simplifile for \$338 million in cash. The cash consideration was gross of \$12 million in cash held by Simplifile on the date of acquisition. Simplifile offers an array of mortgage services, primarily serving as an electronic liaison between lenders, settlement agents and county recording offices, streamlining the local recording of residential mortgage transactions. The transaction expands the ICE Mortgage Services portfolio, which includes MERS. Simplifile is included in our Trading and Clearing segment.

During the nine months ended September 30, 2019, Bakkt acquired two other companies which are not material to our operations.

Regulation

Our markets are primarily subject to the jurisdiction of regulatory agencies in the U.S., U.K., EU, Canada and Singapore. Global policy makers have undertaken reviews of their existing legal framework governing financial markets in connection with regulatory reform, and have either passed new laws and regulations, or are in the process of debating and/or enacting new laws and regulations that apply to our business and to our customers' businesses. Legislative and regulatory actions may impact the way in which we or our customers conduct business and may create uncertainty, which could affect trading volumes or demand for market data. See Part 1, Item 1 “Business - Regulation” and Part 1, Item 1(A) “Risk Factors”

included in our 2018 Form 10-K for a discussion of the primary regulations applicable to our business and certain risks associated with those regulations. As discussed in Part 1, Item 1 of our 2018 Form 10-K, the implementation of the Markets in Financial Instruments Directive II, or MiFID II, and its counterpart the European Market Infrastructure Regulation, or EMIR, may result in operational, regulatory and/or business risk.

Most of the specific regulations which support the MiFID II framework have now been approved or deferred by the European Parliament and European Council. The European Securities and Markets Authority, or ESMA, and the national regulatory authorities are continuing to work on detailed aspects of the framework that are relevant to the markets operated by ICE Futures Europe and ICE Endex, including position limits and the determination of pre-trade price transparency parameters for financial instruments.

The key areas in the evolving regulatory landscape that are likely to impact our business are:

- **The proposed revisions to the regulatory structure of non-EU clearing houses.** On March 13, 2019, the European Parliament, Council and the European Commission reached a political agreement on revisions to the EU's current regulatory and supervisory structure for EU and non-EU clearing houses, called EMIR 2.2. These proposed revisions of the regulatory structure could have an impact on our non-EU clearing houses if they are determined to be systemically important or likely to become systemically important to the financial stability of the EU or one or more of its Member States. It remains uncertain what the nature and extent of the regulation's impact will be on the regulation and supervision of one or more of our non-EU clearing houses, which will depend on ESMA's future determination of whether a non-EU clearing house is systemically important to the EU or its Member States and the extent to which ESMA will rely on such clearing house's domestic regulator. The exact date of the application of EMIR 2.2 is currently unclear, but could be as early as the first quarter of 2020.
- **Basel III capital charges.** The implementation of capital charges in Basel III, particularly, the Supplemental Leverage Ratio applicable to certain financial institutions⁽¹⁾, may impose capital requirements on certain of our clearing house members and their customers that may raise the costs and thus discourage financial institutions from client clearing. The Federal Reserve Board, the Federal Deposit Insurance Corporation and the Office of the Comptroller of the Currency have proposed rule changes to the derivative exposure calculations and leverage ratio requirements; however these regulations could have a negative impact on certain of our clearing members.
- **Capital requirements for investment firms acting as market makers.** EU policy makers are developing a framework for European investment firms. The proposed rules risk imposing disproportionate capital requirements on European investment firms acting as market makers. European investment firms may be discouraged from acting as market makers on certain markets operated by ICE Futures Europe and ICE Endex due to the increased capital requirements.
- **Continued access by EU and U.K. market participants to CCPs and exchanges.** The U.K. and EU continue to negotiate the exit of the U.K. from the EU (commonly known as Brexit). The EU has recently agreed to an extension of the date of the U.K.'s exit from the EU to allow for the U.K.'s ratification of the withdrawal agreement. The extension will last until the sooner to occur of January 31, 2020 or the date of the U.K.'s ratification of the withdrawal agreement. The U.K. government plans to hold a general election on December 12, 2019 and has indicated that it will not progress the ratification of the withdrawal agreement until after the general election takes place. Therefore, the status of the ratification of the withdrawal agreement will depend upon the outcome of the U.K. general election. In anticipation of Brexit, the European Commission adopted an equivalence decision regarding the U.K.'s legal and supervisory arrangements for CCPs in December 2018, and in February 2019, ESMA approved ICE Clear Europe as a third-country CCP. Upon the U.K.'s exit from the EU, EU market participants will be able to continue clearing through U.K. CCPs, such as ICE Clear Europe, until March 31, 2020 in the case of a U.K. exit from the EU without a transition period. Separately, following Brexit, ICE Futures Europe and ICE Endex will continue to be able to permit access by persons in the EU and the U.K., respectively, to trading on their platforms, even in the absence of a transition agreement. However, the lack of an equivalence decision may result in increased costs for certain EU and U.K. market participants which could impact trading on ICE Futures Europe and ICE Endex. The impact to our business and corresponding regulatory changes remain uncertain at this time. We are monitoring the impact to our business as a result of these discussions and are pursuing avenues to facilitate continued access for EU customers to our services in the event the U.K. exits the EU without a transition period.

⁽¹⁾ In June 2019, the Basel Committee on Banking Supervision revised its treatment of the leverage ratio capital requirement for derivatives that a bank centrally clears on behalf of its clients. The revised treatment will permit both cash and non-cash forms of initial margin and variation margin received from a client to offset the replacement cost and potential future exposure for client cleared derivatives only. The revision will apply to the version of the leverage ratio standard that will serve as the Pillar 1 minimum capital requirement as of January 1, 2022.

- **The SEC Transaction Fee Pilot.** In December 2018, the SEC adopted a Transaction Fee Pilot. The final rule establishes a pilot program, for at least one-year and up to two-years, that will limit the fees charged and rebates paid by our five national securities exchanges in certain securities to be designated by the SEC. On March 28, 2019, the SEC partially stayed the Transaction Fee Pilot, pending resolution of our and two other national securities exchanges' petitions for review of the Transaction Fee Pilot by the U.S. Court of Appeals for the District of Columbia Circuit. The only requirement that the SEC did not stay was the requirement that the exchanges collect data during a pre-pilot period, which began on July 1, 2019 and terminates on December 31, 2019.
- **The EU Benchmark Regulation, or BMR.** In June 2016, the EU Benchmark Regulation, or BMR, was adopted and applies from January 2018. Under the BMR, benchmarks provided by a third-country benchmark administrator may be used by EU-supervised entities provided that the European Commission has adopted an equivalence decision or the administrator has been recognized or endorsed and the benchmarks are listed on the register established by ESMA. The BMR provides for a transition period which was extended by the EU authorities to January 1, 2022 for providers of critical benchmarks and third-country benchmark providers. In May 2019, ICE Data Indices, LLC received recognition from the U.K. Financial Conduct Authority, and as such, benchmarks provided by ICE Data Indices, LLC and included in the ESMA register may continue to be used by supervised entities.
- **MiFID II Pre-Trade Price Transparency Requirements.** In June 2019, ESMA issued a Supervisory Briefing to the national regulatory authorities of EU Member States that sets out a supervisory action plan for the pre-trade price transparency regime for commodity derivatives. The supervisory action plan calls upon national regulatory authorities to ensure that the trading venues they regulate either operate under a MiFID II compliant pre-trade transparency waiver or have implemented a full pre-trade transparency regime by December 31, 2019. These requirements could have an impact on the trading arrangements for certain commodity contracts listed on ICE Futures Europe and ICE Endex.

Consolidated Financial Highlights

The following summarizes our results and significant changes in our consolidated financial performance for the periods presented (dollars in millions, except per share amounts and YTD represents the nine-month periods ended September 30th):

REVENUES, LESS TRANSACTION-BASED EXPENSES



Trading and Clearing Segment
Data and Listings Segment

OPERATING INCOME



Trading and Clearing Segment
Data and Listings Segment

ADJUSTED OPERATING INCOME (1)



Trading and Clearing Segment
Data and Listings Segment

CASH FLOWS FROM OPERATIONS



NET INCOME ATTRIBUTABLE TO ICE



ADJUSTED NET INCOME ATTRIBUTABLE TO ICE (1)



	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Revenues, less transaction-based expenses	\$ 3,904	\$ 3,671	6 %	\$ 1,336	\$ 1,200	11 %
Operating expenses	\$ 1,853	\$ 1,764	5 %	\$ 630	\$ 598	5 %
Adjusted operating expenses ⁽¹⁾	\$ 1,619	\$ 1,518	7 %	\$ 551	\$ 521	6 %
Operating income	\$ 2,051	\$ 1,907	8 %	\$ 706	\$ 602	17 %
Adjusted operating income ⁽¹⁾	\$ 2,285	\$ 2,153	6%	\$ 785	\$ 679	16%
Operating margin	53%	52%	1 pt	53%	50%	3 pts
Adjusted operating margin ⁽¹⁾	59%	59%	—	59%	57%	2 pts
Other income (expense), net	\$ (157)	\$ (125)	26 %	\$ (66)	\$ (48)	37 %
Income tax expense	\$ 387	\$ 381	2 %	\$ 103	\$ 89	17 %
Effective tax rate	20%	21%	(1 pt)	16%	16%	—
Net income attributable to ICE	\$ 1,485	\$ 1,377	8 %	\$ 529	\$ 458	16 %
Adjusted net income attributable to ICE ⁽¹⁾	\$ 1,660	\$ 1,541	8 %	\$ 599	\$ 491	22%
Diluted earnings per share attributable to ICE common stockholders	\$ 2.62	\$ 2.37	11 %	\$ 0.94	\$ 0.79	19 %
Adjusted diluted earnings per share attributable to ICE common stockholders ⁽¹⁾	\$ 2.93	\$ 2.65	11 %	\$ 1.06	\$ 0.85	25 %
Cash flows from operating activities	\$ 1,882	\$ 1,735	8 %			

⁽¹⁾ The adjusted figures exclude items that are not reflective of our ongoing core operations and business performance. Adjusted net income attributable to ICE and adjusted diluted earnings per share attributable to ICE common stockholders, are presented net of taxes. These adjusted numbers are not calculated in accordance with GAAP. See “Non-GAAP Financial Measures” below.

- Revenues, less transaction-based expenses, increased \$233 million and \$136 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. See "-Trading and Clearing Segment" and "Data and Listings Segment" below for a discussion of the significant changes in our revenues. The increase in revenues during the nine and three months ended September 30, 2019 includes \$33 million and \$10 million, respectively, in unfavorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2018. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.
- Operating expenses increased \$89 million and \$32 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. See "-Consolidated Operating Expenses" below for a discussion of the significant changes in our operating expenses. The increase in operating expenses during the nine and three months ended September 30, 2019 includes \$14 million and \$4 million, respectively, in favorable foreign exchange effects arising from fluctuations in the U.S. dollar from the comparable periods in 2018. See Item 3 "Quantitative and Qualitative Disclosures About Market Risk-Foreign Currency Exchange Rate Risk" below for additional information on the impact of currency fluctuations.

Variability in Quarterly Comparisons

Our business environment has been characterized by:

- globalization of marketplaces, customers and competitors;
- commodity, interest rate and financial markets uncertainty;
- growing demand for data to inform customers' risk management and investment decisions;
- evolving and disparate regulation across multiple jurisdictions;
- price volatility increasing customers' demand for risk management services;
- increasing focus on capital and cost efficiencies;
- customers' preference to manage risk in markets demonstrating the greatest depth of liquidity and product diversity;
- the evolution of existing products and new product innovation to uniquely serve emerging customer needs;
- rising demand for speed, data, data capacity and connectivity by market participants, necessitating increased investment in technology; and
- consolidation and increasing competition among global markets for trading, clearing and listings.

For additional information regarding the factors that affect our results of operations, see Item 1(A) "Risk Factors" included in our 2018 Form 10-K.

Segment Results

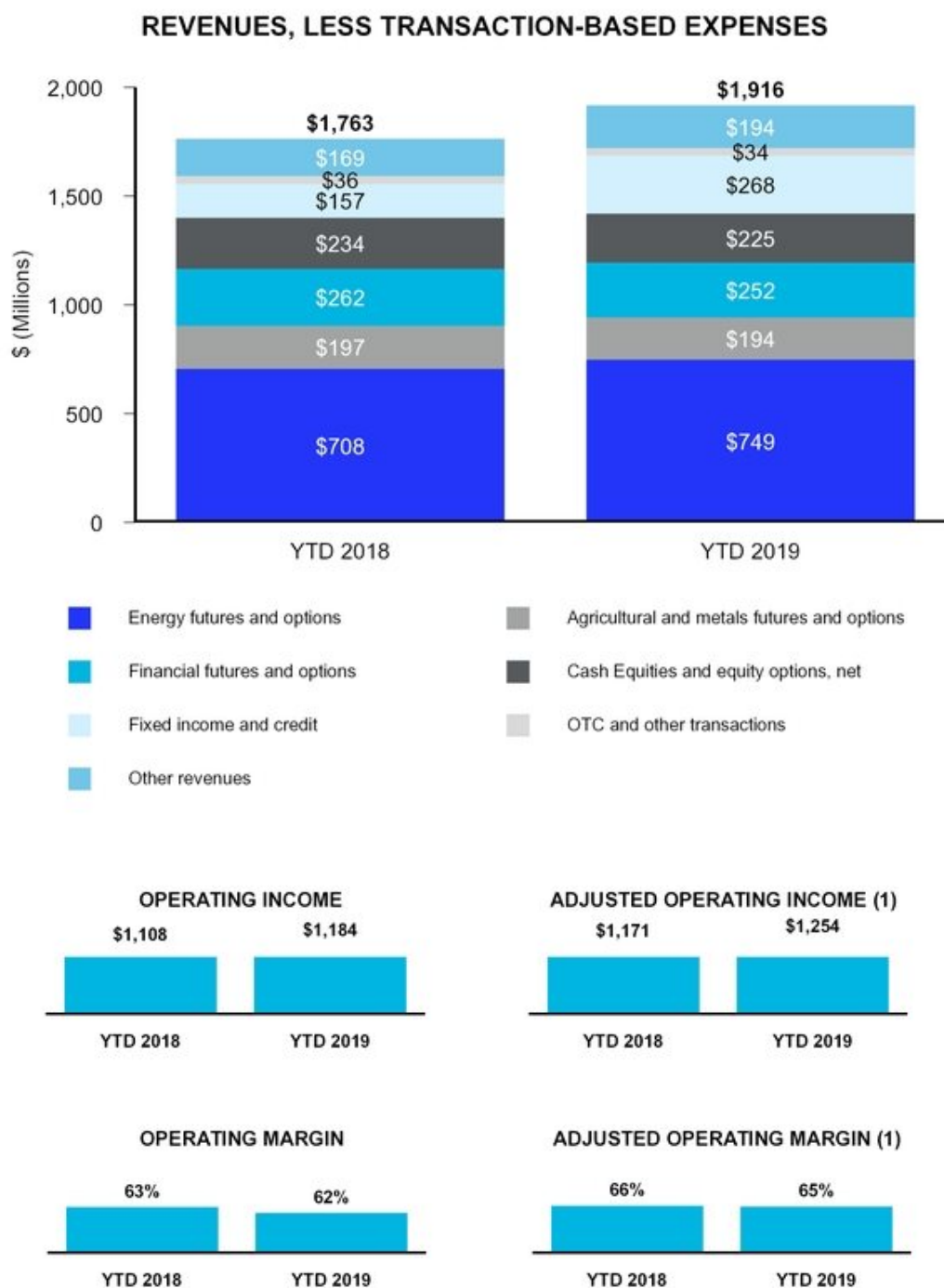
Our business is conducted through two reportable business segments:

- Trading and Clearing, which comprises our transaction-based execution and clearing businesses; and
- Data and Listings, which comprises our subscription-based data services and securities listings businesses.

While revenues are recorded specifically in the segment in which they are earned or to which they relate, a significant portion of our operating expenses are not solely related to a specific segment because the expenses serve functions that are necessary for the operation of both segments. We use a pro-rata revenue approach as the allocation method for the expenses that do not relate solely to one segment. Further, we did not allocate expenses to specific revenue streams within these segments since such an allocation is not reasonably possible. Our two segments do not engage in intersegment transactions.

Trading and Clearing Segment

The following presents selected statements of income data for our Trading and Clearing segment (dollars in millions and YTD represents the nine-month periods ended September 30th):



⁽¹⁾ The adjusted numbers in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Revenues:						
Energy futures and options contracts	\$ 749	\$ 708	6 %	\$ 265	\$ 223	19 %
Agricultural and metals futures and options contracts	194	197	(1)	60	58	4
Financial futures and options contracts	252	262	(4)	91	77	18
Cash equities and equity options	1,201	1,162	3	401	335	20
Fixed income and credit	268	157	70	101	56	81
OTC and other transactions	34	36	(8)	11	11	(3)
Transaction and clearing, net	2,698	2,522	7	929	760	22
Other revenues	194	169	15	67	61	9
Revenues	2,892	2,691	7	996	821	21
Transaction-based expenses	976	928	5	327	263	24
Revenues, less transaction-based expenses	1,916	1,763	9	669	558	20
Other operating expenses	554	492	13	195	171	14
Depreciation and amortization	177	155	14	60	53	13
Acquisition-related transaction and integration costs	1	8	(83)	—	6	(99)
Operating expenses	732	655	12	255	230	11
Operating income	\$ 1,184	\$ 1,108	7 %	\$ 414	\$ 328	26 %

Transaction and Clearing Revenues

Our transaction and clearing revenues are reported on a net basis, except for the NYSE transaction-based expenses discussed below, and consist of fees collected from our derivatives, fixed income, cash equities and equity options trading, derivatives clearing and mortgage and technology services. In our derivatives markets, we earn transaction and clearing revenues from both counterparties to each contract that is traded and/or cleared, and in our equity and equity options markets, we receive trade execution fees as well as routing fees related to orders in our markets which are routed to other markets for execution.

Rates per-contract, or RPC, are driven by the number of contracts or securities traded and the fees charged per contract, net of certain rebates. Our per-contract transaction and clearing revenues will depend upon many factors, including, but not limited to, market conditions, transaction and clearing volume, product mix, pricing, applicable revenue sharing and market making agreements, and new product introductions. Because transaction and clearing revenues are generally assessed on a per-contract basis, revenues and profitability fluctuate with changes in contract volume and due to product mix.

For both the nine months ended September 30, 2019 and 2018, 20% of our Trading and Clearing segment revenues, less transaction-based expenses, were billed in pounds sterling or euros and for both the three months ended September 30, 2019 and 2018, 19% were billed in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues denominated in foreign currencies changes accordingly. For the nine and three months ended September 30, 2019, foreign currency fluctuations of the U.S. dollar as compared to the pound sterling and euro resulted in a decrease to our Trading and Clearing segment revenues, less transaction-based expenses, of \$23 million and \$7 million, respectively, from the comparable periods in 2018. See Item 3 “- Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” below for additional information on the impact of currency fluctuations.

Our transaction and clearing revenues are presented net of rebates. We recorded rebates of \$653 million and \$615 million for the nine months ended September 30, 2019 and 2018, respectively, and \$218 million and \$183 million for the three months ended September 30, 2019 and 2018, respectively. We offer rebates in certain of our markets primarily to support market liquidity and trading volume by providing qualified participants in those markets a discount to the applicable commission rate. Such rebates are calculated based on volumes traded.

- **Energy Futures and Options Contracts:** Total energy volume decreased 1% and revenues increased 6%, respectively, for the nine months ended September 30, 2019 from the comparable period in 2018. Total energy volume increased 14% and revenues increased 19% for the three months ended September 30, 2019 from the comparable period in 2018. Volumes decreased for the nine months ended September 30, 2019 due to lower crude oil and North American natural gas volumes, while revenues increased during the nine months ended September 30, 2019 due to an increase in EU natural gas products and other crude and refined oil products that generally have a higher rate per

contract. Volume and revenues increased during the three months ended September 30, 2019 due to higher oil price volatility and continued strength in our European natural gas complex.

Total oil volume increased 1% for the nine months ended September 30, 2019, from the comparable period in 2018 in part due to a confluence of various geopolitical events and uncertainties as well as supply and demand dynamics. Total oil volume increased 17% for the three months ended September 30, 2019, from the comparable period in 2018 due to higher oil price volatility driven by geopolitical events.

Our global natural gas futures and options volume decreased 5% and increased 10% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. The volume decrease for the nine months ended September 30, 2019 was primarily due to depressed natural gas prices driven by a continued surplus of natural gas in the U.S., specifically related to the Henry Hub contract. The decrease in our Henry Hub natural gas volume was partially offset by increased volumes in our European TTF gas contract. As natural gas continues to globalize, TTF is emerging as the European benchmark for natural gas. Global natural gas volume increased during the three months ended September 30, 2019 due to higher price volatility than the prior year period and continued strength in our European natural gas complex. Emissions futures and options volumes increased 13% and were flat for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. The increase for the nine months ended September 30, 2019 was driven by higher carbon prices and supply-demand dynamics impacted by regulatory uncertainty.

- **Agricultural and Metals Futures and Options Contracts:** Total volume increased 2% and revenues decreased 1% in our agricultural and metals futures and options markets for the nine months ended September 30, 2019 from the comparable period in 2018. Total volume increased 7% and revenues increased 4% in our agricultural and metals futures and options markets for the three months ended September 30, 2019 from the comparable period in 2018. Volume in our largest agricultural contract, sugar futures and options, increased 4% and 7% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. Other agricultural and metal futures and options volume increased 1% and 7% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. The overall increase in agricultural volumes for the nine and three months ended September 30, 2019 was primarily driven by price volatility resulting from supply and demand dynamics, weather concerns and geopolitical events.
- **Financial Futures and Options Contracts:** Interest rate futures and options volume and revenue decreased 7% and 11%, respectively, for the nine months ended September 30, 2019 from the comparable period in 2018. Interest rate futures and options volume and revenue increased 14% and 13%, respectively, for the three months ended September 30, 2019 from the comparable period in 2018. Our interest rate futures and options volume decreased for the nine months ended September 30, 2019, due in part, to a muted European economic backdrop. Interest rate futures and options volume increased for the three months ended September 30, 2019 due to geopolitical events in the U.K. Interest rate futures and options revenues were \$152 million and \$170 million for the nine months ended September 30, 2019 and 2018, respectively, and \$55 million and \$48 million for the three months ended September 30, 2019 and 2018, respectively.

Other financial futures and options volume, which includes our MSCI®, FTSE® and NYSE® FANG+™ equity index products, increased 1% and 19% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018 primarily due to heightened volatility as compared to the prior year periods. Other financial futures and options revenues were \$100 million and \$92 million for the nine months ended September 30, 2019 and 2018, respectively, and \$36 million and \$29 million for the three months ended September 30, 2019 and 2018, respectively.

- **Cash Equities and Equity Options:** Cash equities handled volume increased 9% and 10% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018 due to increased market share and heightened volatility as compared to the prior year period. Cash equities revenues, net of transaction-based expenses, were \$153 million and \$157 million for the nine months ended September 30, 2019 and 2018, respectively, and \$51 million and \$47 million for the three months ended September 30, 2019 and 2018, respectively.

Equity options volume decreased 1% for the nine months ended September 30, 2019 driven by lower industry volumes as compared to the same prior year period. Equity options volume was flat for the three months ended September 30, 2019 primarily due to higher industry volumes, offset by lower market share as compared to the same prior year period. Equity options revenues, net of transaction-based expenses, were \$72 million and \$77 million for the nine months ended September 30, 2019 and 2018, respectively, and \$23 million and \$25 million for the three months ended September 30, 2019 and 2018, respectively.

- **Fixed Income and Credit:** Fixed income and credit includes an increase in fixed income transaction revenues of \$115 million and \$42 million for the nine and three months ended September 30, 2019, respectively, due to acquisitions made in our ICE Mortgage Services and ICE Bonds businesses in late 2018 and in 2019.

CDS clearing revenues were \$102 million and \$105 million for the nine months ended September 30, 2019 and 2018, respectively, and \$35 million and \$32 million for the three months ended September 30, 2019 and 2018, respectively. The notional value of CDS cleared was \$11.9 trillion and \$12.1 trillion, for the nine months ended September 30, 2019 and 2018, respectively, and \$4.3 trillion and \$3.6 trillion for the three months ended September 30, 2019 and 2018, respectively. The decrease in CDS clearing revenues for the nine months ended September 30, 2019 from the comparable period in 2018, was driven by lower market volatility, partially offset by continued record levels of buy-side participation at our U.S. CDS clearing house, ICE Clear Credit, in terms of the number of participants and single name notional cleared. The increase in CDS clearing revenues for the three months ended September 30, 2019 from the comparable period in 2018, was driven by continued record levels of buy-side participation at ICE Clear Credit in terms of the number of participants and single name notional cleared. Fixed income and credit also includes revenues from ICE Mortgage Services and ICE Bonds.

- **OTC and Other Transactions:** OTC and other transactions include revenues from our OTC energy business and other trade confirmation services.
- **Other Revenues:** Other revenues primarily include interest income on certain clearing margin deposits, regulatory penalties and fines, fees for use of our facilities, regulatory fees charged to member organizations of our U.S. securities exchanges, designated market maker service fees, exchange membership fees and agricultural grading and certification fees. The increase in other revenues for the nine and three months ended September 30, 2019 from the comparable periods in 2018 is primarily due to increased interest income earned on certain clearing margin deposits reflecting higher balances and increased interest rates as compared to the same prior year periods. The increase was partially offset by a \$5 million settlement received by NYSE Regulation from ING Financial Markets, LLC, or ING, in September 2018 following an investigation into improper conduct by ING's lending desk from 2007 to 2015.

Selected Operating Data

The following charts and tables present trading activity in our futures and options markets by commodity type based on the total number of contracts traded, as well as futures and options rate per contract (in millions, except for percentages and rate per contract amounts and YTD represents the nine-month periods ended September 30th):

Volume and Rate per Contract

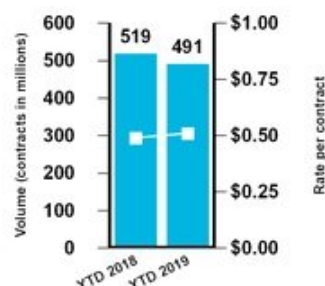
ENERGY FUTURES AND OPTIONS



AGRICULTURAL AND METALS FUTURES AND OPTIONS



FINANCIAL FUTURES AND OPTIONS



Number of contracts traded (in millions):

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Energy futures and options	502	507	(1)%	177	155	14 %
Agricultural and metals futures and options	85	83	2	26	25	7
Financial futures and options	491	519	(6)	174	150	15
Total	1,078	1,109	(3)%	377	330	14 %

Average Daily Volume of contracts traded (in thousands):

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Energy futures and options	2,671	2,695	(1)%	2,762	2,456	12 %
Agricultural and metals futures and options	453	442	2	414	393	5
Financial futures and options	2,565	2,715	(6)	2,650	2,336	13
Total	5,689	5,852	(3)%	5,826	5,185	12 %

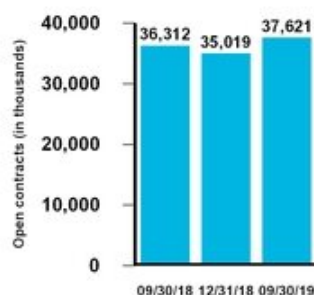
Rate per contract:

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Energy futures and options	\$ 1.49	\$ 1.40	7 %	\$ 1.50	\$ 1.44	4 %
Agricultural and metals futures and options	\$ 2.28	\$ 2.37	(4)%	\$ 2.27	\$ 2.34	(3)%
Financial futures and options	\$ 0.51	\$ 0.49	3 %	\$ 0.52	\$ 0.50	3 %

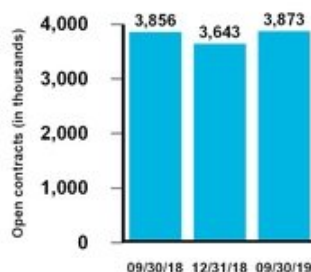
Open interest is the aggregate number of contracts (long or short) that clearing members hold either for their own account or on behalf of their clients. Open interest refers to the total number of contracts that are currently "open," – in other words, contracts that have been traded but not yet liquidated by either an offsetting trade, exercise, expiration or assignment. Open interest is also a measure of the future activity remaining to be closed out in terms of the number of contracts that members and their clients continue to hold in the particular contract and by the number of contracts held for each contract month listed by the exchange. The following charts and table present our quarter-end open interest for our futures and options contracts (in thousands, except for percentages):

Open Interest

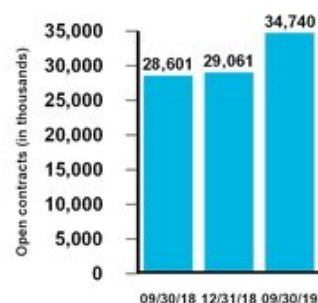
ENERGY FUTURES AND OPTIONS



AGRICULTURAL AND METALS FUTURES AND OPTIONS



FINANCIAL FUTURES AND OPTIONS

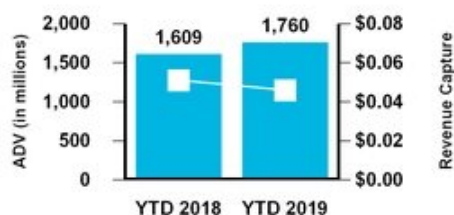


Open interest — in thousands of contracts:

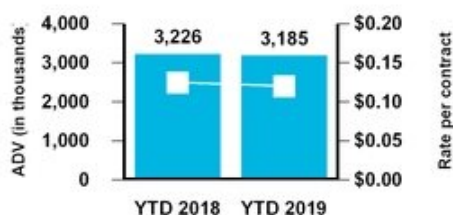
	As of September 30,		Change
	2019	2018	
Energy futures and options	37,621	36,312	4%
Agricultural and metals futures and options	3,873	3,856	—
Financial futures and options	34,740	28,601	21
Total	76,234	68,769	11%

The following charts and table present selected cash and equity options trading data (all trading volume below is presented as average net daily trading volume, or ADV, and is single counted and YTD represents the nine-month periods ended September 30th):

NYSE CASH EQUITIES



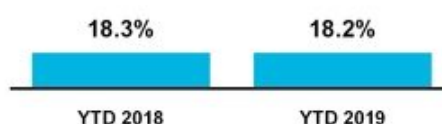
NYSE EQUITY OPTIONS



NYSE EQUITY MARKET SHARE (MATCHED)



NYSE EQUITY OPTIONS MARKET SHARE



	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
NYSE cash equities (shares in millions):						
NYSE listed (Tape A) issues:						
Handled volume	1,145	1,095	5 %	1,095	985	11 %
Matched volume	1,133	1,087	4 %	1,083	976	11 %
Total NYSE listed consolidated volume	3,600	3,500	3 %	3,527	3,144	12 %
Share of total matched consolidated volume	31.5%	31.0%	0.4 pts	30.7%	31.0%	(0.4) pts
NYSE Arca, NYSE American and regional listed (Tape B) issues:						
Handled volume	371	324	14 %	359	308	17 %
Matched volume	362	315	15 %	351	301	17 %
Total NYSE Arca, NYSE American and regional listed consolidated volume	1,362	1,293	5 %	1,364	1,149	19 %
Share of total matched consolidated volume	26.6%	24.4%	2.2 pts	25.7%	26.2%	(0.4) pts
Nasdaq listed (Tape C) issues:						
Handled volume	244	190	29 %	211	219	(4)%
Matched volume	230	178	29 %	198	206	(4)%
Total Nasdaq listed consolidated volume	2,178	2,156	1 %	2,047	2,067	(1)%
Share of total matched consolidated volume	10.6%	8.3%	2.3 pts	9.7%	9.9%	(0.3) pts
Total cash handled volume	1,760	1,609	9 %	1,665	1,512	10 %
Total cash market share matched	24.2%	22.7%	1.4 pts	23.5%	23.3%	0.2 pts
NYSE equity options (contracts in thousands):						
NYSE equity options volume	3,185	3,226	(1)%	3,062	3,070	— %
Total equity options volume	17,478	17,657	(1)%	17,767	16,505	8 %
NYSE share of total equity options	18.2%	18.3%	—	17.2%	18.6%	(1.4) pts
Revenue capture or rate per contract:						
Cash equities rate per contract (per 100 shares)	\$0.046	\$0.051	(10)%	\$0.049	\$0.048	— %
Equity options rate per contract	\$0.12	\$0.13	(4)%	\$0.12	\$0.12	(5)%

Handled volume represents the total number of shares of equity securities, ETFs and crossing session activity internally matched on our exchanges or routed to and executed on an external market center. Matched volume represents the total number of shares of equity securities, ETFs and crossing session activity executed on our exchanges.

Transaction-Based Expenses

Our equities and equity options markets pay fees to the SEC pursuant to Section 31 of the Exchange Act. Section 31 fees are recorded on a gross basis as a component of transaction and clearing fee revenue. These Section 31 fees are assessed to recover the government's costs of supervising and regulating the securities markets and professionals and are subject to change. We, in turn, collect corresponding activity assessment fees from member organizations clearing or settling trades on the equities and options exchanges, and recognize these amounts in our transaction and clearing revenues when invoiced. The activity assessment fees are designed to equal the Section 31 fees. As a result, activity assessment fees and the corresponding Section 31 fees do not have an impact on our net income, although the timing of payment by us may vary from collections. Section 31 fees were \$274 million and \$272 million for the nine months ended September 30, 2019 and 2018, respectively, and \$105 million and \$61 million for the three months ended September 30, 2019 and 2018, respectively. The fees we collect are included in cash at the time of receipt and we remit the amounts to the SEC semi-annually as required. The total amount is included in accrued liabilities and was \$34 million as of September 30, 2019.

We make liquidity payments to cash and options trading customers, as well as routing charges made to other exchanges which are included in transaction-based expenses. We incur routing charges when we do not have the best bid or offer in the market for a security that a customer is trying to buy or sell on one of our securities exchanges. In that case, we route the customer's order to the external market center that displays the best bid or offer. The external market center charges us a fee per share (denominated in tenths of a cent per share) for routing to its system. We record routing charges on a gross basis as a component of transaction and clearing fee revenue. Cash liquidity payments, routing and clearing fees were \$702 million and \$656 million for the nine months ended September 30, 2019 and 2018, respectively, and \$222 million and \$202 million for the three months ended September 30, 2019 and 2018, respectively.

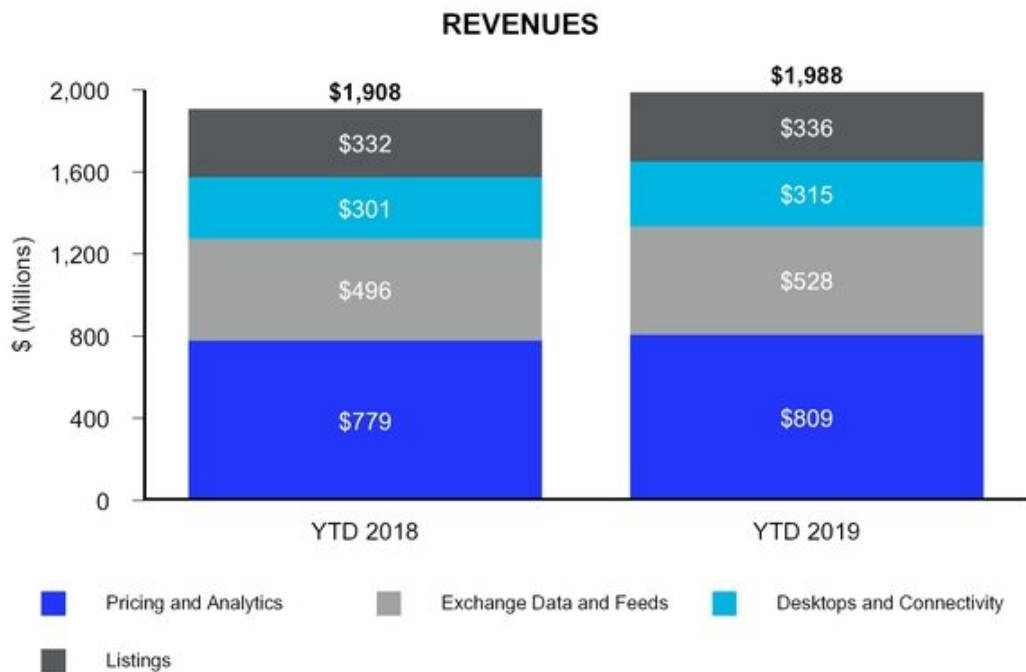
Operating Expenses, Operating Income and Operating Margin

Our Trading and Clearing segment operating expenses increased \$77 million and \$25 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018, largely related to our acquisitions of CHX Holdings, Inc., or CHX, TMC Bonds and MERS in 2018 and Simplifile in 2019 and our launch of Bakkt. See “- Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses. Our Trading and Clearing segment operating income increased \$76 million and \$86 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. Trading and Clearing segment operating margins were 62% and 63% for the nine months ended September 30, 2019 and 2018, respectively, and 62% and 59% for the three months ended September 30, 2019 and 2018, respectively.

Our Trading and Clearing segment adjusted operating expenses were \$662 million and \$592 million for the nine months ended September 30, 2019 and 2018, respectively, and \$231 million and \$206 million for the three months ended September 30, 2019 and 2018, respectively. Our Trading and Clearing segment adjusted operating income was \$1.3 billion and \$1.2 billion for the nine months ended September 30, 2019 and 2018, respectively, and \$438 million and \$352 million for the three months ended September 30, 2019 and 2018, respectively. Our Trading and Clearing segment adjusted operating margins were 65% and 66% for the nine months ended September 30, 2019 and 2018, respectively, and 65% and 63% for the three months ended September 30, 2019 and 2018, respectively. See “- Non-GAAP Financial Measures” below.

Data and Listings Segment

The following charts and table present our selected statements of income data for our Data and Listings segment (dollars in millions and YTD represents the nine-month periods ended September 30th):





(1) The adjusted numbers in the charts above are calculated by excluding items that are not reflective of our cash operations and core business performance. As a result, these adjusted numbers are not calculated in accordance with U.S. GAAP. See “- Non-GAAP Financial Measures” below.

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Revenues:						
Pricing and analytics	\$ 809	\$ 779	4 %	\$ 273	\$ 263	4 %
Exchange data and feeds	528	496	6	172	168	3
Desktops and connectivity	315	301	5	108	99	9
Data services	1,652	1,576	5	553	530	4
Listings	336	332	1	114	112	2
Revenues	1,988	1,908	4	667	642	4
Other operating expenses	825	810	2	277	273	1
Depreciation and amortization	296	274	8	98	95	3
Acquisition-related transaction and integration costs	—	25	(100)	—	—	(100)
Operating expenses	1,121	1,109	1	375	368	2
Operating income	\$ 867	\$ 799	9 %	\$ 292	\$ 274	7 %

Data Services Revenues

Our Data and Listings segment represents largely subscription-based, or recurring, revenues from data services and listings services offered across our trading and clearing businesses and ICE Data Services. Through ICE Data Services, we generate revenues from a range of global financial and commodity markets, including pricing and reference data, exchange data, analytics, feeds, index services, desktops and connectivity solutions. Through NYSE, NYSE American and NYSE Arca, we generate listings revenue related to the provision of listings services for public companies and ETFs, and related corporate actions for listed companies.

For both the nine months ended September 30, 2019 and 2018, 7% of our Data and Listings segment revenues were billed in pounds sterling or euros and for both the three months ended September 30, 2019 and 2018, 7% were billed in pounds sterling or euros (all relating to our data services revenues). As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues denominated in foreign currencies changes accordingly. Due to fluctuations in the U.S. dollar compared to the pound sterling and euro, our Data and Listings segment revenues were \$9 million lower for the nine months ended September 30, 2019, from the comparable period in 2018 and \$3 million lower for the three months ended September 30, 2019, from the comparable period in 2018.

Our data services revenues are primarily subscription-based and increased 5% and 4% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. The increase in revenues for the nine and

three months ended September 30, 2019 was primarily due to the strong retention rate of existing customers, the addition of new customers, increased purchases by existing customers and increases in pricing of our products.

- **Pricing and Analytics:** Our pricing and analytics revenues increased 4% for both the nine and three months ended September 30, 2019, from the comparable periods in 2018. The increase in revenue was driven by the strong retention rate of existing customers, the addition of new customers, increased purchases by existing customers and increases in pricing of our products. This growth was somewhat offset by unfavorable fluctuations in the U.S. dollar compared to the pound sterling and euro.
- **Exchange Data and Feeds:** Our exchange data and feeds revenues increased 6% and 3% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. The increase in revenue was driven by the strong retention rate of existing customers, the addition of new customers, increased purchases by existing customers, a larger share of the National Market System, or NMS, Plan revenue and increases in pricing of our products.
- **Desktops and Connectivity:** Our desktop and connectivity revenues increased 5% and 9% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018 driven primarily by stronger desktop revenues as well as an increase in revenue from connectivity services including the ICE Global Network. This growth was somewhat offset by unfavorable fluctuations in the U.S. dollar compared to the pound sterling and euro.

Annual Subscription Value, or ASV, represents, at a point in time, the data services revenues subscribed for the succeeding 12 months. ASV does not include new sales, contract terminations or price changes that may occur during that 12-month period. ASV also does not include certain data services revenue streams that are not subscription-based. Revenue from ASV businesses has historically represented approximately 90% of our data revenues. Thus, while it is an indicative forward-looking metric, it does not provide a growth forecast of the next 12 months of data services revenues.

As of September 30, 2019, ASV was \$1.995 billion, which increased 4.9% compared to the ASV as of September 30, 2018. This does not adjust for year-over-year foreign exchange fluctuations or impacts of acquisitions.

Listings Revenues

Listings revenues in our securities markets arise from fees applicable to companies listed on our cash equities exchanges — original listing fees and annual listing fees. Original listing fees consist of two components: initial listing fees and fees related to corporate actions. Initial listing fees, subject to a minimum and maximum amount, are based on the number of shares that a company initially lists. All listings fees are billed upfront and the identified performance obligations are satisfied over time. Revenue related to the investor relations performance obligation is recognized ratably over a two-year period, with the remaining revenue recognized ratably over time as customers continue to list on our exchanges, which is generally estimated to be over a period of up to nine years for NYSE and up to five years for NYSE Arca and NYSE American.

In addition, we earn corporate actions-related listing fees in connection with actions involving the issuance of new shares, such as stock splits, rights issues and sales of additional securities, as well as mergers and acquisitions. Listings fees related to other corporate actions are considered contract modifications of our listing contracts and are recognized ratably over time as customers continue to list on our exchanges, which is generally estimated to be a period of six years for NYSE and three years for NYSE Arca and NYSE American.

Our listings revenues increased 1% and 2% for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018 due to changes in market conditions for IPOs.

Operating Expenses, Operating Income and Operating Margin

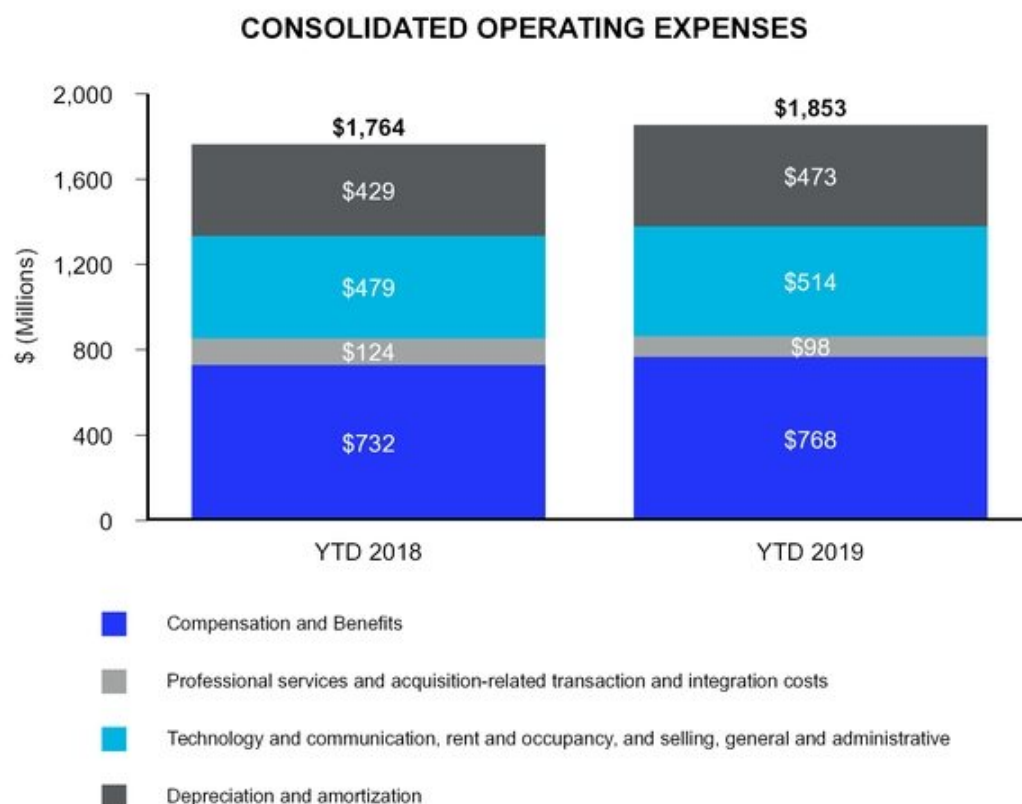
Our Data and Listings segment operating expenses increased \$12 million and \$7 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. See “- Consolidated Operating Expenses” below for a discussion of the significant changes in our operating expenses. Our Data and Listings segment operating income increased \$68 million and \$18 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. Our Data and Listings segment operating margins were 44% and 42% for the nine months ended September 30, 2019 and 2018, respectively, and 44% and 43% for the three months ended September 30, 2019 and 2018, respectively. The operating income and operating margin increases were driven by the revenue increases discussed above.

Our Data and Listings segment adjusted operating expenses were \$957 million and \$926 million for the nine months ended September 30, 2019 and 2018, respectively, and \$320 million and \$315 million for the three months ended September 30, 2019 and 2018, respectively. Our Data and Listings segment adjusted operating income was \$1.0 billion and \$982 million

for the nine months ended September 30, 2019 and 2018, respectively, and \$347 million and \$327 million for the three months ended September 30, 2019 and 2018, respectively. Our Data and Listings segment adjusted operating margins were 52% and 51% for the nine months ended September 30, 2019 and 2018, respectively, and 52% and 51% for the three months ended September 30, 2019 and 2018, respectively. See “— Non-GAAP Financial Measures” below.

Consolidated Operating Expenses

The following presents our consolidated operating expenses (dollars in millions and YTD represents the nine-month periods ended September 30th):



	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Compensation and benefits	\$ 768	\$ 732	5 %	\$ 261	\$ 251	4 %
Professional services	97	91	6	35	32	7
Acquisition-related transaction and integration costs	1	33	(96)	—	6	(99)
Technology and communication	346	320	8	126	107	18
Rent and occupancy	52	50	3	17	17	1
Selling, general and administrative	116	109	6	33	37	(11)
Depreciation and amortization	473	429	10	158	148	7
Total operating expenses	\$ 1,853	\$ 1,764	5 %	\$ 630	\$ 598	5 %

The majority of our operating expenses do not vary directly with changes in our volume and revenues, except for certain technology and communication expenses, including data acquisition costs, licensing and other fee-related arrangements and a portion of our compensation expense that is tied directly to our data sales or overall financial performance.

We expect our operating expenses to increase in absolute terms in future periods in connection with the growth of our business, and to vary from year-to-year based on the type and level of our acquisitions, our integrations and other investments.

For the nine months ended September 30, 2019 and 2018, 12% and 13%, respectively, of our operating expenses were incurred in pounds sterling or euros and for the three months ended September 30, 2019 and 2018, 12% and 13%, respectively, were billed in pounds sterling or euros. Due to fluctuations in the U.S. dollar compared to the pound sterling and euro, our consolidated operating expenses decreased \$14 million for the nine months ended September 30, 2019 and decreased \$4 million for the three months ended September 30, 2019, from the comparable periods in 2018. See Item 3 “— Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” below for additional information.

Compensation and Benefits Expenses

Compensation and benefits expense is our most significant operating expense and includes non-capitalized employee wages, bonuses, non-cash or stock compensation, certain severance costs, benefits and employer taxes. The bonus component of our compensation and benefits expense is based on both our financial performance and the individual employee performance. The performance-based restricted stock compensation expense is also based on our financial performance. Therefore, our compensation and benefits expense will vary year-to-year based on our financial performance and fluctuations in the number of employees.

As of September 30, 2019, we had 5,435 employees compared to 5,042 as of September 30, 2018. Our employee headcount increased primarily due to new employees related to Bakkt and the acquisitions of MERS in October 2018 and Simplifile in June 2019. These acquisitions, as well as those of CHX and TMC Bonds in 2018 and the launch of Bakkt contributed to an increase in our compensation and benefits expense of \$41 million and \$7 million for the nine and three months ended September 30, 2019, respectively, from the comparable periods in 2018. Excluding these acquisitions, we incurred employee severance and retention costs of \$13 million and \$23 million for the nine months ended September 30, 2019 and 2018, respectively, and \$3 million and \$6 million for the three months ended September 30, 2019 and 2018, respectively.

Non-cash compensation expenses recognized in our consolidated financial statements for employee stock options and restricted stock were \$100 million and \$93 million for the nine months ended September 30, 2019 and 2018, respectively, and \$36 million and \$32 million for the three months ended September 30, 2019 and 2018, respectively.

Professional Services Expenses

Professional services expense includes fees for consulting services received on strategic and technology initiatives, temporary labor, as well as regulatory, legal and accounting fees, and may fluctuate as a result of changes in consulting and technology services, temporary labor, and regulatory, accounting and legal proceedings.

Professional services expenses increased for the nine and three months ended September 30, 2019 from the comparable periods in 2018 primarily due to legal fees associated with Brexit, regulatory and legal matters relating to the NYSE markets, litigation matters including the class action lawsuits in which the company is a defendant and consulting work related to strategic initiatives.

Acquisition-Related Transaction and Integration Costs

We incurred \$1 million in acquisition-related transaction and integration costs for the nine months ended September 30, 2019. For the nine and three months ended September 30, 2018, we incurred \$33 million and \$6 million, respectively, in acquisition-related transaction and integration costs, primarily relating to employee terminations, lease terminations and professional services costs from our 2018 acquisitions, as well as a \$5 million banker success fee in connection with our acquisition of TMC Bonds. The integration of Interactive Data was completed by June 30, 2018.

We expect to continue to explore and pursue various potential acquisitions and other strategic opportunities to strengthen our competitive position and support our growth. As a result, we may incur acquisition-related transaction costs in future periods. See “- Non-GAAP Financial Measures” below.

Technology and Communication Expenses

Technology support services consist of costs for running our wholly-owned data centers, hosting costs paid to third-party data centers and maintenance of our computer hardware and software required to support our technology and cybersecurity. These costs are driven by system capacity, functionality and redundancy requirements. Communication expenses consist of costs for network connections for our electronic platforms, telecommunications costs, and fees paid for access to external market data. This expense also includes licensing and other fee agreement expenses, which may be impacted by growth in electronic contract volume, our capacity requirements, changes in the number of telecommunications hubs and connections with customers to access our electronic platforms directly.

Technology and communications expenses increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018, primarily due to \$13 million and \$4 million, respectively, in costs related to our acquisitions of CHX, TMC Bonds and MERS in 2018 and Simplifile in 2019 and our launch of Bakkt, and an increase of \$24 million and \$15 million, respectively, from revenue share agreements related to our product offering. Beginning in the second quarter of 2019, we have reflected amounts owed under certain third-party revenue share arrangements as technology and communication operating expenses rather than as had been previously recorded net within transaction and clearing revenues. The increase for the nine months ended September 30, 2019 was partially offset by additional costs related to data migrations and hardware support upgrades of \$11 million during the nine months ended September 30, 2018.

Rent and Occupancy Expenses

Rent and occupancy expense relates to leased and owned property and includes rent, maintenance, real estate taxes, utilities and other related costs. We have significant operations located in and around Atlanta, New York and London with smaller offices located throughout the world.

Rent and occupancy expenses increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018 primarily due to increased costs related to our 2018 acquisitions of CHX, TMC Bonds and MERS.

Selling, General and Administrative Expenses

Selling, general and administrative expenses include marketing, advertising, public relations, insurance, bank service charges, dues and subscriptions, travel and entertainment, non-income taxes and other general and administrative costs.

Selling, general and administrative expenses increased for the nine months ended September 30, 2019, from the comparable period in 2018, primarily due to \$3 million in costs related to our 2018 and 2019 acquisitions of CHX, TMC Bonds, MERS and Simplifile, as well as travel and entertainment and other general and administrative costs, partially offset by the release of non-income tax reserves. Selling, general and administrative expenses decreased for the three months ended September 30, 2019, from the comparable period in 2018 primarily due to the release of non-income tax reserves.

Depreciation and Amortization Expenses

Depreciation and amortization expense results from depreciation of long-lived assets such as buildings, leasehold improvements, aircraft, hardware and networking equipment, software, furniture, fixtures and equipment over their estimated useful lives. This expense includes amortization of intangible assets obtained in our acquisitions of businesses, as well as on various licensing agreements, over their estimated useful lives. Intangible assets subject to amortization consist primarily of customer relationships, trading products with finite lives and technology. This expense also includes amortization of internally-developed and purchased software over its estimated useful life.

We recorded amortization expenses on intangible assets acquired as part of our acquisitions, as well as on other intangible assets, of \$235 million and \$211 million for the nine months ended September 30, 2019 and 2018, respectively, and \$79 million and \$73 million for the three months ended September 30, 2019 and 2018, respectively. In addition, for the nine months ended September 30, 2018, we recorded a \$4 million impairment loss on exchange registration intangible assets related to our closure of ICE Futures Canada and ICE Clear Canada as amortization expense. Amortization expense increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018 as a result of CHX, TMC Bonds and MERS intangible assets.

We recorded depreciation expenses on our fixed assets of \$238 million and \$214 million for the nine months ended September 30, 2019 and 2018, respectively, and \$79 million and \$75 million for the three months ended September 30, 2019 and 2018, respectively. Depreciation expense increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018 primarily due to depreciation resulting from increased software development and networking equipment.

Consolidated Non-Operating Income (Expense)

Income and expenses incurred through activities outside of our core operations are considered non-operating. The following table presents our non-operating income (expense) (dollars in millions):

	Nine Months Ended September 30,			Three Months Ended September 30,		
	2019	2018	Change	2019	2018	Change
Other income (expense):						
Interest income	\$ 27	\$ 15	79 %	\$ 8	\$ 6	36 %
Interest expense	(214)	(173)	24	(72)	(66)	10
Other income, net	30	33	(10)%	(2)	12	n/a
Total other income (expense), net	\$ (157)	\$ (125)	26 %	\$ (66)	\$ (48)	37 %
Net income attributable to non-controlling interest	\$ (22)	\$ (24)	(12)%	\$ (8)	\$ (7)	(2)%

Interest Income

Interest income increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018 primarily due to a rise in short-term interest rates on various investments.

Interest Expense

Interest expense increased for the nine and three months ended September 30, 2019, from the comparable periods in 2018 primarily due to an increase in the principal and coupon of our bond refinancing in August 2018, as well as a rise in short-term interest rates impacting our Commercial Paper Program.

Other income, net

In connection with our equity investment in Euroclear, we recognized dividend income of \$15 million during the nine months ended September 30, 2018, which is included in other income. No Euroclear dividends were declared during the nine months ended September 30, 2019. We expect a Euroclear dividend to be declared and received during the three months ended December 31, 2019.

During the nine and three months ended September 30, 2019, we recorded promissory note impairment charges of \$16 million on work performed by the original plan processor on the consolidated audit trail, or CAT. In 2016, the SEC approved a plan to establish a market-wide CAT to improve regulators' ability to monitor trading activity with a phased implementation through 2019. Funding of the implementation and operation of the CAT was ultimately expected to be provided by both the execution venues, including us, and industry members. To date, however, funding has been provided solely by the execution venues in exchange for promissory notes expected to be repaid if industry member fees are: (i) approved by the SEC and (ii) collected to fund the CAT. Due to delays and failures in implementation and functionality by the original plan processor, as well as recently published proposals by the SEC for an amended timeline and implementation structure, we believe the risk that execution venues are not reimbursed has increased, resulting in this impairment.

Our equity method investments include the OCC, and prior to purchasing the remaining minority stake in MERS in October 2018, our majority investment in MERS. We recognized \$51 million and \$27 million in equity income related to these investments as other income for the nine months ended September 30, 2019 and 2018, respectively, and \$15 million and \$13 million for the three months ended September 30, 2019 and 2018, respectively.

Prior to October 2018, we owned a majority stake in MERS and treated it as an equity method investment because we did not have the ability to control its operations. On October 3, 2018, we completed the purchase of all remaining interests of MERS. On that date, we ceased recording equity income of MERS. MERS is now part of ICE Mortgage Services.

We own a 40% interest in the OCC, which is regulated by the SEC and the CFTC. On February 13, 2019, the SEC disapproved the OCC capital plan that was established in 2015. Following the SEC disapproval, the OCC also announced they will not be providing a refund to clearing members or declaring a dividend to shareholders for the year ended December 31, 2018, which resulted in higher reported OCC 2018 net income than we had estimated. During the nine and three months ended September 30, 2019, we recognized \$51 million and \$15 million, respectively, of equity earnings as our share of estimated OCC profits, including \$19 million related to 2018 earnings which was recognized during the nine months ended September 30, 2019. During the nine and three months ended September 30, 2018, we recognized \$12 million and \$6 million, respectively, of equity earnings on our OCC investment. Refer to Note 4 to our consolidated financial statements, included in this Form 10-Q for additional details on our OCC investment.

We incurred foreign currency transaction losses of \$1 million for both the nine months ended September 30, 2019 and 2018, primarily attributable to fluctuations of the pound sterling and euro relative to the U.S. dollar. Foreign currency transaction gains and losses are recorded in other income, net, when the settlement of foreign currency assets, liabilities and payables that occur through our foreign operations that are received in non-functional currencies due to the increase or decrease in the period-end foreign currency exchange rates between periods. See Item 3 “- Quantitative and Qualitative Disclosures About Market Risk - Foreign Currency Exchange Rate Risk” included elsewhere in this Quarterly Report for more information on these items.

Non-controlling Interest

For consolidated subsidiaries in which our ownership is less than 100% and for which we have control over the assets and liabilities and the management of the entity, the outside stockholders' interests are shown as non-controlling interests. Non-controlling interests include those related to the operating results of our CDS clearing subsidiaries in which non-ICE limited partners hold a 26.7% ownership interest as of September 30, 2019.

In December 2018, Bakkt was capitalized with \$183 million in initial funding with ICE as the majority owner, along with a group of other minority investors. We hold a call option over these interests subject to certain terms. Similarly, the non-ICE partners in Bakkt hold a put option to require us to repurchase their interests subject to certain terms. These minority interests are reflected as redeemable non-controlling interests in temporary equity within our consolidated balance sheet.

Consolidated Income Tax Provision

Consolidated income tax expense was \$387 million and \$381 million for the nine months ended September 30, 2019 and 2018, respectively, and \$103 million and \$89 million for the three months ended September 30, 2019 and 2018, respectively. The change in consolidated income tax expense between periods is primarily due to the tax impact of changes in our pre-tax income and the changes in our effective tax rate each period. Our effective tax rate was 20% and 21% for the nine months ended September 30, 2019 and 2018, respectively, and 16% for both the three months ended September 30, 2019 and 2018. The effective tax rate for the nine months ended September 30, 2019 is lower than the effective tax rate for the same period in 2018 primarily due to changes in estimates of certain international tax provisions enacted as part of the TCJA and reductions of income tax reserves resulting from the expiration of certain statutes of limitation.

Quarterly Results of Operations

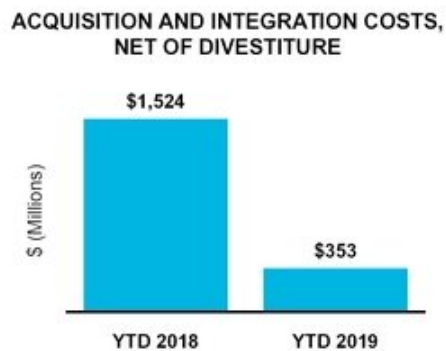
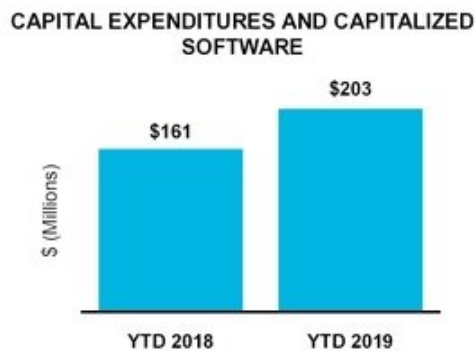
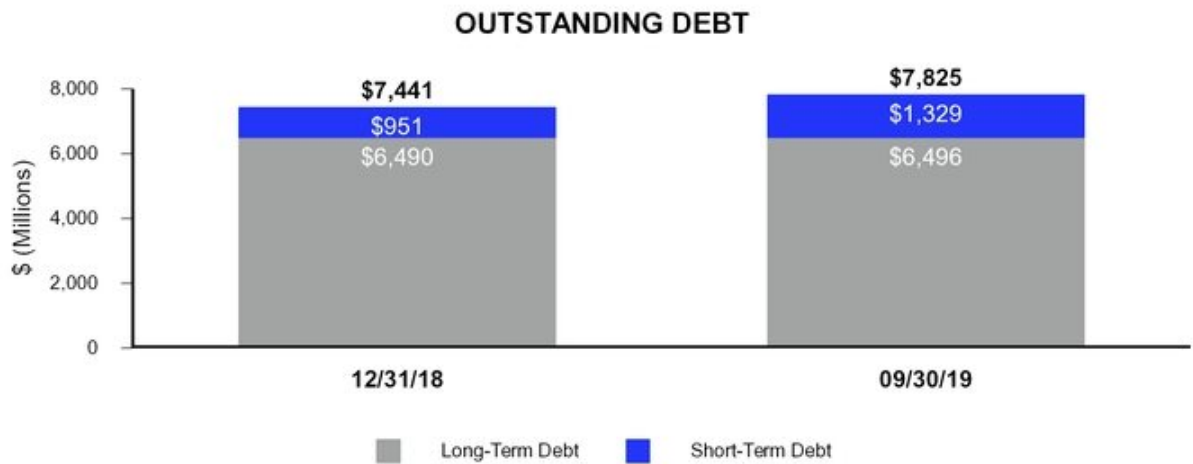
The following quarterly unaudited condensed consolidated statements of income data has been prepared on substantially the same basis as our audited consolidated financial statements and includes all adjustments, consisting only of normal recurring adjustments, necessary for the fair presentation of our consolidated results of operations for the quarters presented. The historical results for any quarter do not necessarily indicate the results expected for any future period. The following table sets forth quarterly consolidated statements of income data (in millions):

	Three Months Ended,				
	September 30, 2019	June 30, 2019	March 31, 2019	December 31, 2018	September 30, 2018
Revenues:					
Energy futures and options contracts	\$ 265	\$ 255	\$ 229	\$ 257	\$ 223
Agricultural and metals futures and options contracts	60	72	62	54	58
Financial futures and options contracts	91	78	83	92	77
Cash equities and equity options	401	410	390	462	335
Fixed income and credit	101	80	87	83	56
OTC and other transactions	11	12	11	13	11
Total transaction and clearing, net	929	907	862	961	760
Pricing and analytics	273	270	266	264	263
Exchange data and feeds	172	180	176	174	168
Desktops and connectivity	108	103	104	101	99
Total data services	553	553	546	539	530
Listings	114	111	111	112	112
Other revenues	67	63	64	65	61
Total revenues	1,663	1,634	1,583	1,677	1,463
Transaction-based expenses	327	336	313	369	263
Total revenues, less transaction-based expenses	1,336	1,298	1,270	1,308	1,200
Compensation and benefits	261	259	248	262	251
Professional services	35	29	33	40	32
Acquisition-related transaction and integration costs	—	1	—	1	6
Technology and communication	126	113	107	112	107
Rent and occupancy	17	18	17	18	17
Selling, general and administrative	33	41	42	42	37
Depreciation and amortization	158	157	158	157	148
Total operating expenses	630	618	605	632	598
Operating income	706	680	665	676	602
Other income (expense), net ⁽¹⁾	(66)	(52)	(39)	62	(48)
Income tax expense	103	150	134	119	89
Net income	\$ 537	\$ 478	\$ 492	\$ 619	\$ 465
Net income attributable to non-controlling interest	(8)	(6)	(8)	(8)	(7)
Net income attributable to Intercontinental Exchange, Inc.	\$ 529	\$ 472	\$ 484	\$ 611	\$ 458

⁽¹⁾ Other income (expense), net for the three months ended December 31, 2018 includes a \$110 million gain in connection with our acquisition of MERS.

Liquidity and Capital Resources

Below are charts that reflect our capital allocation. The acquisition and integration costs in the chart below includes cash paid for acquisitions, net of cash received for divestitures, cash paid for equity investments, cash paid for non-controlling interest and redeemable non-controlling interest, and the acquisition-related transaction and integration costs, net of divestitures, in each period (YTD represents the nine-month periods ended September 30th).



We have financed our operations, growth and cash needs primarily through income from operations and borrowings under our various debt facilities. Our principal capital requirements have been to fund capital expenditures, working capital, strategic acquisitions and investments, stock repurchases, dividends and the development of our technology platforms. We believe that our cash on hand and cash flows from operations will be sufficient to repay our outstanding debt, but we may also need to incur additional debt or issue additional equity securities in the future. See “- Future Capital Requirements” below.

Our Commercial Paper Program enables us to borrow efficiently at reasonable short-term interest rates and provides us with the flexibility to de-lever using our strong annual cash flows from operating activities whenever our leverage becomes elevated as a result of investment or acquisition activities. We had net issuances of \$367 million under our Commercial Paper Program during the nine months ended September 30, 2019.

Upon maturity of our commercial paper and to the extent old issuances are not repaid by cash on hand, we are exposed to the rollover risk of not being able to issue new commercial paper. To mitigate this risk, we maintain an undrawn back-stop bank revolving credit facility for an aggregate amount which meets or exceeds the amount issued under our Commercial Paper Program at any time. If we were not able to issue new commercial paper, we have the option of drawing on the back-stop revolving facility. However, electing to do so would result in higher interest expense. For a discussion of our Commercial Paper Program and other indebtedness, see “- Debt” below.

Consolidated cash and cash equivalents were \$655 million and \$724 million as of September 30, 2019 and December 31, 2018, respectively. We had \$1.3 billion and \$1.1 billion in short-term and long-term restricted cash and cash equivalents as of September 30, 2019 and December 31, 2018, respectively. We consider all short-term, highly-liquid investments with original maturity dates of three months or less to be cash equivalents. We classify all investments with original maturity dates in excess of three months but less than one year as short-term investments and all investments that we intend to hold for more than one year as long-term investments. Short-term and long-term investments are included in other current and other non-current assets, respectively. Cash and cash equivalents that are not available for general use, either due to regulatory requirements or through restrictions in specific agreements, are classified as restricted cash and cash equivalents and investments.

As of September 30, 2019, the amount of unrestricted cash held by our non-U.S. subsidiaries was \$269 million.

Our cash and cash equivalents and financial investments are managed as a global treasury portfolio of non-speculative financial instruments that are readily convertible into cash, such as overnight deposits, term deposits, money market funds, mutual funds for treasury investments, short duration fixed income investments and other money market instruments, thus ensuring high liquidity of financial assets. We may invest a portion of our cash in excess of short-term operating needs in investment-grade marketable debt securities, including government or government-sponsored agencies and corporate debt securities. As of September 30, 2019, we hold \$49 million of unrestricted cash that is set aside for legal, regulatory, and surveillance operations at NYSE.

In September 2018, our Board of Directors approved an aggregate of \$2.0 billion for future repurchases of our common stock with no fixed expiration date that became effective January 1, 2019. We expect this authorization to provide us with capacity for buybacks over six quarters and flexibility to act opportunistically. We expect funding for any share repurchases to come from our operating cash flow or borrowings under our Commercial Paper Program or our debt facilities.

Repurchases of our common stock may be made from time to time on the open market, through established trading plans, in privately-negotiated transactions or otherwise, in accordance with all applicable securities laws, rules and regulations. For the nine months ended September 30, 2019, we repurchased 12.5 million shares of our outstanding common stock at a cost of \$1.0 billion under our Rule 10b5-1 trading plan and 1.3 million shares at a cost of \$100 million on the open market. For the nine months ended September 30, 2018, we repurchased 14.4 million shares of our outstanding common stock at a cost of \$1.1 billion. The shares repurchased are held in treasury stock.

From time to time, we enter into Rule 10b5-1 trading plans, as authorized by our Board of Directors, to govern some or all of the repurchases of our shares of common stock. The timing and extent of future repurchases that are not made pursuant to a Rule 10b5-1 trading plan will be at our discretion and will depend upon many conditions. In making a determination regarding any stock repurchases, management considers multiple factors, including overall stock market conditions, our common stock price movements, the remaining amount authorized for repurchases by our Board of Directors, the potential impact of a stock repurchase program on our corporate debt ratings, our expected free cash flow and working capital needs, our current and future planned strategic growth initiatives and other potential uses of our cash and capital resources.

We may discontinue stock repurchases at any time and may amend or terminate a Rule 10b5-1 trading plan at any time. The approval of our Board of Directors for the share repurchases does not obligate us to acquire any particular amount of our common stock. In addition, our Board of Directors may increase or decrease the amount of capacity we have for repurchases from time to time.

Cash Flow

The following table presents the major components of net changes in cash, cash equivalents, and restricted cash and cash equivalents (in millions):

	Nine Months Ended September 30,	
	2019	2018
Net cash provided by (used in):		
Operating activities	\$ 1,882	\$ 1,735
Investing activities	(512)	(1,617)
Financing activities	(1,251)	(19)
Effect of exchange rate changes	(2)	(5)
Net increase in cash, cash equivalents and restricted cash and cash equivalents	\$ 117	\$ 94

Operating Activities

Net cash provided by operating activities primarily consists of net income adjusted for certain non-cash items, including depreciation and amortization and the effects of changes in working capital.

The \$147 million increase in net cash provided by operating activities during the nine months ended September 30, 2019 as compared to the prior period in 2018 is primarily a result of an increase in our net income, fluctuations in our working capital and the timing of various payments such as transaction-related expenses and taxes.

Investing Activities

Consolidated net cash used in investing activities for the nine months ended September 30, 2019 relates to \$352 million cash paid for acquisitions (primarily Simplifile), \$87 million of capital expenditures, \$116 million of capitalized software development expenditures and a \$44 million return of capital related to our equity method investment in the OCC. Refer to Note 4 to our consolidated financial statements, included in this Form 10-Q for additional details on our OCC investment.

Consolidated net cash used in investing activities for the nine months ended September 30, 2018 primarily relates to our January 2018 acquisition of BondPoint for \$400 million, the purchase of an additional 5.1% stake in Euroclear for €246 million in cash (\$305 million), our July 2018 acquisition of TMC Bonds for \$701 million, \$49 million of capital expenditures and \$112 million of capitalized software development expenditures.

The capital expenditures primarily relate to hardware and software purchases to continue the development and expansion of our electronic platforms, data services and clearing houses, and leasehold improvements associated with the new and renovated office spaces in Atlanta, New York and London. The software development expenditures primarily relate to the continued development and expansion of our electronic trading platforms, data services and clearing houses.

Financing Activities

Consolidated net cash used in financing activities for the nine months ended September 30, 2019 primarily relates to \$1.1 billion in repurchases of common stock, \$367 million in net borrowings under our Commercial Paper Program, \$467 million in dividend payments to stockholders and \$64 million in cash payments related to treasury shares received for restricted stock tax payments and stock options exercises.

Consolidated net cash used in financing activities for the nine months ended September 30, 2018 primarily relates to \$1.1 billion in repurchases of common stock, \$600 million in repayments of the October 2018 Senior Notes, \$417 million in dividend payments to our stockholders, \$35 million in net repayments under our Commercial Paper Program and \$78 million in cash payments related to treasury shares received for restricted stock tax payments and stock options exercises, partially offset by \$2.2 billion in net proceeds from our September 2023 senior notes, our 2028 senior notes and our 2048 senior notes offering.

Debt

Our total debt, including short-term and long-term debt, consisted of the following:

	As of September 30, 2019	As of December 31, 2018
	(in millions)	
Debt:		
Short-term debt:		
Commercial Paper	\$ 1,318	\$ 951
Other short-term debt	11	—
Total short-term debt	1,329	951
Long-term debt:		
2020 Senior Notes (2.75% senior unsecured notes due December 1, 2020)	1,247	1,246
2022 Senior Notes (2.35% senior unsecured notes due September 15, 2022)	497	496
2023 Senior Notes (3.45% senior unsecured notes due September 21, 2023)	397	397
2023 Senior Notes (4.00% senior unsecured notes due October 15, 2023)	794	793
2025 Senior Notes (3.75% senior unsecured notes due December 1, 2025)	1,244	1,243
2027 Senior Notes (3.10% senior unsecured notes due September 15, 2027)	496	496
2028 Senior Notes (3.75% senior unsecured notes due September 21, 2028)	592	591
2048 Senior Notes (4.25% senior unsecured notes due September 21, 2048)	1,229	1,228
Total long-term debt	6,496	6,490
Total debt	\$ 7,825	\$ 7,441

Credit Facilities

We have a \$3.4 billion senior unsecured revolving Credit Facility with a maturity date of August 9, 2023. The Credit Facility includes an option for us to propose an increase in the aggregate amount available for borrowing by up to \$975 million, subject to the consent of the lenders funding the increase and certain other conditions. No amounts were outstanding under the Credit Facility as of September 30, 2019.

As of September 30, 2019, of the \$3.4 billion that is currently available for borrowing under the Credit Facility, \$1.3 billion is required to back-stop the amount outstanding under our Commercial Paper Program and \$160 million is required to support certain broker-dealer subsidiary commitments. The amount required to back-stop the amounts outstanding under the Commercial Paper Program will fluctuate as we increase or decrease our commercial paper borrowings. The remaining \$1.9 billion is available for working capital and general corporate purposes including, but not limited to, acting as a back-stop to future increases in the amounts outstanding under the Commercial Paper Program.

During the nine months ended September 30, 2019, a subsidiary of ours entered into a new \$20 million line of credit for their general corporate purposes. As of September 30, 2019, the subsidiary had borrowed \$11 million, which is reflected as “other short-term debt” in the table above.

Commercial Paper Program

Our Commercial Paper Program is currently backed by the borrowing capacity available under the Credit Facility, as described above. The effective interest rate of commercial paper issuances does not materially differ from short-term interest rates (such as USD LIBOR) which fluctuate due to market conditions and as a result may impact our interest expense. During the nine months ended September 30, 2019, we had net issuances of \$367 million under the Commercial Paper Program, the proceeds of which were used to fund the acquisition of Simplifile and for general corporate purposes.

Commercial paper notes of \$1.3 billion with original maturities ranging from one to 77 days were outstanding as of September 30, 2019 under our Commercial Paper Program, with a weighted average interest rate of 2.2% per annum and a weighted average maturity of 27 days.

Future Capital Requirements

Our future capital requirements will depend on many factors, including the rate of growth across our Trading and Clearing and Data and Listings segments, strategic plans and acquisitions, available sources for financing activities, required and discretionary technology and clearing initiatives, regulatory requirements, the timing and introduction of new products and enhancements to existing products, the geographic mix of our business and potential stock repurchases.

We currently expect to incur capital expenditures (including operational and real estate capital expenditures) and to incur software development costs that are eligible for capitalization ranging in the aggregate between \$315 million and \$320 million in 2019, which we believe will support the enhancement of our technology, business integration and the continued growth of our businesses.

Effective January 1, 2019, our Board of Directors approved an aggregate of \$2.0 billion for repurchases of our common stock with no fixed expiration date. We expect this authorization to provide us with capacity for buybacks over six quarters and flexibility to act opportunistically. As of September 30, 2019, we had \$880 million authorized for future repurchases of our common stock. Refer to Note 10 to our consolidated financial statements, included in this Form 10-Q for additional details on our stock repurchase plan.

Our Board of Directors has adopted a quarterly dividend policy providing that dividends will be approved quarterly by the board or its Audit Committee taking into account factors such as our evolving business model, prevailing business conditions and our financial results and capital requirements, without a predetermined net income payout ratio. During the third quarter of 2019, we paid a quarterly dividend of \$0.275 per share of our common stock for an aggregate payout of \$155 million, which includes the payment of dividend equivalents on unvested employee restricted stock units. On October 31, 2019, we announced a \$0.275 per share dividend for the fourth quarter of 2019 with the dividend payable on December 31, 2019 to stockholders of record as of December 16, 2019.

As of September 30, 2019, we had \$7.8 billion in outstanding debt. We currently also have a \$3.4 billion Credit Facility, of which \$1.9 billion is available for working capital and general corporate purposes. Other than the facilities discussed above for the ICE Clearing Houses, the Credit Facility and our Commercial Paper Program are currently the only significant agreements or arrangements that we have for liquidity and capital resources with third parties. In the event of any strategic acquisitions, mergers or investments, or if we are required to raise capital for any reason or desire to return capital to our stockholders, we may incur additional debt, issue additional equity to raise necessary funds, repurchase additional shares of our common stock or pay a dividend. However, we cannot provide assurance that such financing or transactions will be available or successful, or that the terms of such financing or transactions will be favorable to us. See "- Debt" above.

Non-GAAP Measures

We use certain financial measures internally to evaluate our performance and make financial and operational decisions that are presented in a manner that adjusts from their equivalent GAAP measures or that supplement the information provided by our GAAP measures. We use these adjusted results because we believe they more clearly highlight trends in our business that may not otherwise be apparent when relying solely on GAAP financial measures, since these measures eliminate from our results specific financial items that have less bearing on our core operating performance.

We use these measures in communicating certain aspects of our results and performance, including in this Quarterly Report, and believe that these measures, when viewed in conjunction with our GAAP results and the accompanying reconciliation, can provide investors with greater transparency and a greater understanding of factors affecting our financial condition and results of operations than GAAP measures alone. In addition, we believe the presentation of these measures is useful to investors for making period-to-period comparisons of results because the adjustments to GAAP are not reflective of our core business performance.

These financial measures are not presented in accordance with, or as an alternative to, GAAP financial measures and may be different from non-GAAP measures used by other companies. We encourage investors to review the GAAP financial measures included in this Quarterly Report, including our consolidated financial statements, to aid in their analysis and understanding of our performance and in making comparisons.

The tables below outline our adjusted operating expenses, adjusted operating income, adjusted operating margin, adjusted net income attributable to ICE common stockholders and adjusted earnings per share, which are non-GAAP measures that are calculated by making adjustments for items we view as not reflective of our cash operations and core business performance. These measures, including the adjustments and their related income tax effect and other tax adjustments (in millions, except for percentages and per share amounts), are as follows:

	Trading and Clearing Segment		Data and Listings Segment		Consolidated	
	Nine Months Ended September 30,		Nine Months Ended September 30,		Nine Months Ended September 30,	
	2019	2018	2019	2018	2019	2018
Total revenues, less transaction-based expenses	\$ 1,916	\$ 1,763	\$ 1,988	\$ 1,908	\$ 3,904	\$ 3,671
Operating expenses	732	655	1,121	1,109	1,853	1,764
Less: Interactive Data integration costs and acquisition-related success fees	—	5	—	24	—	29
Less: Amortization of acquisition-related intangibles	70	50	164	159	234	209
Less: Impairment of exchange registration intangible assets on closure of ICE Futures Canada and ICE Clear Canada	—	4	—	—	—	4
Less: Employee severance costs related to ICE Futures Canada and ICE Clear Canada operations	—	4	—	—	—	4
Adjusted operating expenses	\$ 662	\$ 592	\$ 957	\$ 926	\$ 1,619	\$ 1,518
Operating income	\$ 1,184	\$ 1,108	\$ 867	\$ 799	\$ 2,051	\$ 1,907
Adjusted operating income	\$ 1,254	\$ 1,171	\$ 1,031	\$ 982	\$ 2,285	\$ 2,153
Operating margin	62%	63%	44%	42%	53%	52%
Adjusted operating margin	65%	66%	52%	51%	59%	59%
Net income attributable to ICE common stockholders					\$ 1,485	\$ 1,377
Add: Interactive Data integration costs and acquisition-related success fees					—	29
Add: Amortization of acquisition-related intangibles					234	209
Add: Impairment of CAT promissory notes					16	—
Add: Adjustment to reduce net gain on Trayport divestiture					—	1
Add: Impairment of exchange registration intangible assets on closure of ICE Futures Canada and ICE Clear Canada					—	4
Add: Employee severance costs related to ICE Futures Canada and ICE Clear Canada operations					—	4
Less: Income tax effect for the above items					(65)	(63)
Less: Deferred tax adjustment from U.S. tax rate reduction					—	(12)
Add/(Less): Deferred tax adjustment on acquisition-related intangibles					(13)	5
Add/(Less): Other tax adjustments					3	(13)
Adjusted net income attributable to ICE common stockholders					\$ 1,660	\$ 1,541
Basic earnings per share attributable to ICE common stockholders					\$ 2.64	\$ 2.39
Diluted earnings per share attributable to ICE common stockholders					\$ 2.62	\$ 2.37
Adjusted basic earnings per share attributable to ICE common stockholders					\$ 2.95	\$ 2.67
Adjusted diluted earnings per share attributable to ICE common stockholders					\$ 2.93	\$ 2.65

	Trading and Clearing Segment		Data and Listings Segment		Consolidated	
	Three Months Ended September 30,		Three Months Ended September 30,		Three Months Ended September 30,	
	2019	2018	2019	2018	2019	2018
Total revenues, less transaction-based expenses	\$ 669	\$ 558	\$ 667	\$ 642	\$ 1,336	\$ 1,200
Operating expenses	255	230	375	368	630	598
Less: Acquisition-related success fees	—	5	—	—	—	5
Less: Amortization of acquisition-related intangibles	24	19	55	53	79	72
Adjusted operating expenses	\$ 231	\$ 206	\$ 320	\$ 315	\$ 551	\$ 521
Operating income	\$ 414	\$ 328	\$ 292	\$ 274	\$ 706	\$ 602
Adjusted operating income	\$ 438	\$ 352	\$ 347	\$ 327	\$ 785	\$ 679
Operating margin	62%	59%	44%	43%	53%	50%
Adjusted operating margin	65%	63%	52%	51%	59%	57%
Net income attributable to ICE common stockholders					\$ 529	\$ 458
Add: Acquisition-related success fees					—	5
Add: Amortization of acquisition-related intangibles					79	72
Add: Impairment of CAT promissory notes					16	—
Less: Income tax effect for the above items					(25)	(19)
Less: Deferred tax adjustment from U.S. tax rate reduction					—	(12)
Less: Other tax adjustments					—	(13)
Adjusted net income attributable to ICE common stockholders					\$ 599	\$ 491
Basic earnings per share attributable to ICE common stockholders					\$ 0.95	\$ 0.80
Diluted earnings per share attributable to ICE common stockholders					\$ 0.94	\$ 0.79
Adjusted basic earnings per share attributable to ICE common stockholders					\$ 1.07	\$ 0.86
Adjusted diluted earnings per share attributable to ICE common stockholders					\$ 1.06	\$ 0.85

Acquisition-related transaction costs are included as part of our core business expenses, except for those that are directly related to the announcement, closing, financing or termination of a transaction. However, we adjusted for the acquisition-related integration costs relating to Interactive Data given the size of this acquisition. The integration of Interactive Data was completed by June 2018. Amortization of acquisition-related intangibles are included in non-GAAP adjustments as excluding these non-cash expenses provides greater clarity regarding our financial strength and stability of cash operating results.

In addition, we include the 2019 promissory note impairment charges on work performed by the original plan processor on the consolidated audit trail, or CAT. In 2016, the SEC approved a plan to establish a market-wide CAT to improve regulators' ability to monitor trading activity with a phased implementation through 2019. Funding of the implementation and operation of the CAT was ultimately expected to be provided by both the execution venues, including us, and industry members. To date, however, funding has been provided solely by the execution venues in exchange for promissory notes expected to be repaid if industry member fees are: (i) approved by the SEC and (ii) collected to fund the CAT. Due to delays and failures in implementation and functionality by the original plan processor, as well as recently published proposals by the SEC for an amended timeline and implementation structure, we believe the risk that execution venues are not reimbursed has increased, resulting in this impairment. This is included as a non-GAAP adjustment as this is not considered a part of our core business operations.

During the nine months ended September 30, 2018, we include as non-GAAP adjustments the 2018 reduction of the gain on our December 2017 sale of Trayport, as this is not considered a part of our core business operations. In addition, we include the impairment loss on exchange registration intangible assets and employee severance costs related to the closure of ICE Futures Canada and ICE Clear Canada as non-GAAP adjustments as they are non-recurring items. The income tax effects relating to all non-GAAP adjustments above, as well as deferred tax adjustments on acquisition-related intangibles are included as non-GAAP adjustments. The \$3 million other tax adjustment for the nine months ended September 30, 2019 is for additional audit settlement payments primarily related to pre-acquisition tax matters in conjunction with our acquisition of NYSE in 2013. During the nine and three months ended September 30, 2018, we include a \$12 million non-GAAP adjustment from deferred tax benefits associated with the U.S. tax rate reduction resulting from changes in estimates. Other tax adjustments during the nine and three months ended September 30, 2018 include

a \$17 million tax benefit on the sale of Trayport offset by an audit settlement for a pre-acquisition period in connection with our acquisition of NYSE in 2013.

For additional information on these items, refer to our consolidated financial statements included in this Quarterly Report and “—Consolidated Operating Expenses” and “—Consolidated Non-Operating Income (Expenses)”, above.

Contractual Obligations and Commercial Commitments

During the nine months ended September 30, 2019, there were no significant changes to our contractual obligations and commercial commitments from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2018 Form 10-K, other than the changes related to our adoption of the new lease accounting standard, ASU 2016-02, described in Note 2 of our consolidated financial statements.

Off-Balance Sheet Arrangements

As described in Note 12 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, certain clearing house collateral is reported off-balance sheet. We do not have any relationships with unconsolidated entities or financial partnerships, often referred to as structured finance or special purpose entities.

New and Recently Adopted Accounting Pronouncements

Refer to Note 2 to our consolidated financial statements, which are included elsewhere in this Quarterly Report, for information on the new and recently adopted accounting pronouncements that are applicable to us.

Critical Accounting Policies

During the nine months ended September 30, 2019, there were no significant changes to our critical accounting policies and estimates from those disclosed in the section “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in our 2018 Form 10-K.

ITEM 3. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

As a result of our operating and financing activities, we are exposed to market risks such as interest rate risk, foreign currency exchange rate risk and credit risk. We have implemented policies and procedures designed to measure, manage, monitor and report risk exposures, which are regularly reviewed by the appropriate management and supervisory bodies.

Interest Rate Risk

We have exposure to market risk for changes in interest rates relating to our cash and cash equivalents, short-term and long-term restricted cash and cash equivalents, short-term and long-term investments and indebtedness. As of September 30, 2019 and December 31, 2018, our cash and cash equivalents, short-term and long-term investments, and short-term and long-term restricted cash and cash equivalents were both \$2.0 billion of which \$250 million and \$275 million, respectively, were denominated in pounds sterling, euros or Canadian dollars. The remaining cash and cash equivalents, short-term and long-term investments, and short-term and long-term restricted cash and cash equivalents are denominated in U.S. dollars. We do not use our investment portfolio for trading or other speculative purposes. A hypothetical 100 basis point decrease in short-term interest rates would decrease annual pre-tax earnings by \$17 million as of September 30, 2019, assuming no change in the amount or composition of our cash and cash equivalents and short-term and long-term restricted cash and cash equivalents.

As of September 30, 2019, we had \$7.8 billion in outstanding debt, of which \$6.5 billion relates to our senior notes, which bear interest at fixed interest rates. The remaining amount outstanding of \$1.3 billion relates to the Commercial Paper Program, which bears interest at fluctuating rates and, therefore, subjects us to interest rate risk, and a subsidiary line of credit. A hypothetical 100 basis point increase in short-term interest rates relating to the amounts outstanding under the Commercial Paper Program as of September 30, 2019 would decrease annual pre-tax earnings by \$13 million, assuming no change in the volume or composition of our outstanding indebtedness and no hedging activity. See Item 2 “Management’s Discussion and Analysis of Financial Condition and Results of Operations - Debt” included in this Quarterly Report.

The interest rates on our Commercial Paper Program are currently evaluated based upon current maturities and market conditions. The weighted average interest rate on our Commercial Paper Program decreased from 2.48% as of December 31, 2018 to 2.20% as of September 30, 2019, and increased from 2.15% as of September 30, 2018. The decrease in the Commercial Paper Program weighted average interest rate in 2019 was primarily due to the decision by the U.S. Federal Reserve in July 2019 to decrease the federal funds short-term interest rate by 25 basis points. The effective interest rate of commercial paper issuances will continue to fluctuate based on the movement in short-term interest rates along with shifts in supply and demand within the commercial paper market.

Foreign Currency Exchange Rate Risk

As an international business, we are subject to foreign currency exchange rate risk. We may experience gains or losses from foreign currency transactions in the future given that a significant part of our assets and liabilities are recorded in pounds sterling, Canadian dollars or euros, and a significant portion of our revenues and expenses are recorded in pounds sterling or euros. Certain assets, liabilities, revenues and expenses of foreign subsidiaries are denominated in the local functional currency of such subsidiaries. Our exposure to foreign denominated earnings for the nine and three months ended September 30, 2019 and September 30, 2018 is presented by primary foreign currency in the following table (dollars in millions, except exchange rates):

	Nine Months Ended September 30, 2019		Three Months Ended September 30, 2019		Nine Months Ended September 30, 2018		Three Months Ended September 30, 2018	
	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro	Pound Sterling	Euro
Average exchange rate to the U.S. dollar in the current year period	\$ 1.2734	\$ 1.1237	\$ 1.2330	\$ 1.1118	\$ 1.3520	\$ 1.1947	\$ 1.3037	\$ 1.1631
Average exchange rate to the U.S. dollar in the same period in the prior year	\$ 1.3520	\$ 1.1947	\$ 1.3037	\$ 1.1631	\$ 1.2760	\$ 1.1136	\$ 1.3091	\$ 1.1752
Average exchange rate increase (decrease)	(6)%	(6)%	(5)%	(4)%	6%	7%	— %	(1)%
Foreign denominated percentage of:								
Revenues, less transaction-based expenses	8 %	5 %	8 %	5 %	9%	5%	8 %	4 %
Operating expenses	10 %	2 %	10 %	2 %	11%	2%	11 %	2 %
Operating income	7 %	8 %	7 %	8 %	7%	7%	6 %	7 %
Impact of the currency fluctuations ⁽¹⁾ on:								
Revenues, less transaction-based expenses	\$ (20)	\$ (13)	\$ (7)	\$ (3)	\$ 20	\$ 11	\$ —	\$ (1)
Operating expenses	\$ (11)	\$ (3)	\$ (3)	\$ (1)	\$ 11	\$ 3	\$ —	\$ —
Operating income	\$ (9)	\$ (10)	\$ (4)	\$ (2)	\$ 9	\$ 8	\$ —	\$ (1)

(1) Represents the impact of currency fluctuation for the nine and three months ended September 30, 2019 and September 30, 2018 compared to the same periods in the prior year.

We have a significant part of our assets, liabilities, revenues and expenses recorded in pounds sterling or euros. For both the nine and three months ended September 30, 2019, 13% of our consolidated revenues, less transaction-based expenses, were denominated in pounds sterling or euros, and for both the nine and three months ended September 30, 2019, 12% of our consolidated operating expenses were denominated in pounds sterling or euros. As the pound sterling or euro exchange rate changes, the U.S. equivalent of revenues and expenses denominated in foreign currencies changes accordingly.

Foreign currency transaction risk related to the settlement of foreign currency denominated assets, liabilities and payables occur through our operations, which are received in or paid in pounds sterling, Canadian dollars or euros, due to the increase or decrease in the foreign currency exchange rates between periods. We incurred foreign currency transaction losses of \$1 million for both the nine months ended September 30, 2019 and 2018. A 10% adverse change in the underlying foreign currency exchange rates as of September 30, 2019, assuming no change in the composition of the foreign currency denominated assets, liabilities and payables and assuming no hedging activity, would result in a foreign currency transaction loss of \$3 million.

We entered into foreign currency hedging transactions during the nine and three months ended September 30, 2019 as economic hedges to help mitigate a portion of our foreign exchange risk exposure and may enter into additional hedging transactions in the future to help mitigate our foreign exchange risk exposure. Although we may enter into additional hedging transactions in the future to help mitigate our foreign exchange risk exposure, these hedging arrangements may not be effective, particularly in the event of imprecise forecasts of the levels of our non-U.S. denominated assets and liabilities.

We have foreign currency translation risk equal to our net investment in our foreign subsidiaries. The financial statements of these subsidiaries are translated into U.S. dollars using a current rate of exchange, with gains or losses included in the

cumulative translation adjustment account, a component of equity. Our exposure to the net investment in foreign currencies is presented by primary foreign currencies in the table below (in millions):

	As of September 30, 2019		
	Position in pounds sterling	Position in Canadian dollars	Position in euros
Assets	£ 825	C\$ 1,249	€ 151
of which goodwill represents	619	409	92
Liabilities	89	831	46
Net currency position	£ 736	C\$ 418	€ 105
Net currency position, in \$USD	\$ 905	\$ 316	\$ 114
Negative impact on consolidated equity of a 10% decrease in foreign currency exchange rates	\$ 90	\$ 32	\$ 11

As of September 30, 2019 and December 31, 2018, the portion of our equity attributable to accumulated other comprehensive loss from foreign currency translation was \$259 million and \$227 million, respectively. For the nine and three months ended September 30, 2019, currency exchange rate differences had a negative impact of \$32 million and \$39 million, respectively, on our consolidated equity. The decrease for the nine and three months ended September 30, 2019 is primarily due to the decrease in the pound sterling/U.S. dollar exchange rate to 1.2295 as of September 30, 2019 (from 1.2756 as of December 31, 2018) due to the weakening of the pound sterling as compared to the U.S. dollar. The future impact on our business relating to the U.K. leaving the EU and the corresponding regulatory changes are uncertain at this time, including future impacts on currency exchange rates.

Credit Risk

We are exposed to credit risk in our operations in the event of a counterparty default. We limit our exposure to credit risk by rigorously selecting the counterparties with which we make our investments, monitoring them on an ongoing basis and executing agreements to protect our interests.

Clearing House Cash Deposit Risks

The ICE Clearing Houses hold material amounts of clearing member cash deposits which are held or invested primarily to provide security of capital while minimizing credit, market and liquidity risks. Refer to Note 12 to our consolidated financial statements, which are included in this Quarterly Report, for more information on the ICE Clearing Houses' cash deposits, which were \$66.0 billion as of September 30, 2019. While we seek to achieve a reasonable rate of return which may generate interest income for our clearing members, we are primarily concerned with preservation of capital and managing the risks associated with these deposits. As the ICE Clearing Houses may pass on interest revenues (minus costs) to the clearing members, this could include negative or reduced yield due to market conditions. For a summary of the risks associated with this investment activity and how these risks are mitigated, see Part II, Item 7(A) "Quantitative and Qualitative Disclosures About Market Risk" in our 2018 Form 10-K.

ITEM 4. CONTROLS AND PROCEDURES

(a) *Evaluation of Disclosure Controls and Procedures.* As of the end of the period covered by this report, an evaluation was carried out by our management, with the participation of our Chief Executive Officer and Chief Financial Officer, of the effectiveness of our disclosure controls and procedures (as defined in Rule 13a-15(e) under the Exchange Act). Based upon that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that these disclosure controls and procedures were effective as of the end of the period covered by this report.

(b) *Changes in Internal Controls over Financial Reporting.* There were no changes in our internal controls over financial reporting that occurred during our most recent fiscal quarter that have materially affected, or are reasonably likely to materially affect, our internal controls over financial reporting. As a result, no corrective actions were taken.

PART II. Other Information

ITEM 1. LEGAL PROCEEDINGS

We are subject to legal proceedings, claims and investigations that arise in the ordinary course of our business. We establish accruals for those matters in circumstances when a loss contingency is considered probable and the related amount is reasonably estimable. Any such accruals may be adjusted as circumstances change. Assessments of losses are inherently subjective and involve unpredictable factors. We do not believe that the resolution of these legal matters will have a material adverse effect on our consolidated financial condition, results of operations or liquidity. It is possible, however, that future results of operations for any particular quarterly or annual period may be materially and adversely affected by any developments in legal proceedings, claims and investigations. See Part I, Item 3 “Legal Proceedings” in our 2018 Form 10-K and Note 13 to the consolidated financial statements and related notes, which are included in this Quarterly Report.

ITEM 1(A). RISK FACTORS

During the nine months ended September 30, 2019, there were no significant new risk factors from those disclosed in Part 1, Item 1A, “Risk Factors” in our 2018 Form 10-K. In addition to the other information set forth in this Quarterly Report, including the information in the “- Regulation” section of Item 2 - Management’s Discussion and Analysis of Financial Condition and Results of Operations, you should carefully consider the factors discussed under “Risk Factors” and the regulation discussion under “Business - Regulation” in our 2018 Form 10-K. These risks could materially and adversely affect our business, financial condition and results of operations. The risks and uncertainties in our 2018 Form 10-K are not the only ones facing us. Additional risks and uncertainties not presently known to us, or that we currently believe to be immaterial, may also adversely affect our business.

ITEM 2. UNREGISTERED SALES OF EQUITY SECURITIES AND USE OF PROCEEDS

Stock Repurchases

The table below sets forth the information with respect to purchases made by or on behalf of ICE or any “affiliated purchaser” (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during the three months ended September 30, 2019.

Period (2019)	Total number of shares purchased (in thousands)	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs (in thousands) (1)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions) (1)
July 1 - July 31	1,306	\$89.61	11,347	\$1,103
August 1 - August 31	1,289	\$91.44	12,636	\$985
September 1 - September 30	1,135	\$92.62	13,771	\$880
Total	3,730	\$91.16	13,771	\$880

We have provided in the table below additional supplemental information with respect to purchases made by or on behalf of ICE or any “affiliated purchaser” (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during each of the three months ended March 31, 2019, June 30, 2019 and September 30, 2019.

Three months ended:	Total number of shares purchased (in thousands)	Average price paid per share	Total number of shares purchased as part of publicly announced plans or programs (in thousands) (1)	Approximate dollar value of shares that may yet be purchased under the plans or programs (in millions) (1)
March 31, 2019	5,871	\$74.95	5,871	\$1,560
June 30, 2019	4,170	\$81.53	10,041	\$1,220
September 30, 2019	3,730	\$91.16	13,771	\$880
Total	13,771	\$81.33	13,771	\$880

(1) Refer to the Notes to our consolidated financial statements and related notes, which are included in our Quarterly Report on Form 10-Q for the quarters ended March 31, 2019 and June 30, 2019 and elsewhere in this Quarterly Report, for details on our stock repurchase plans.

ITEM 3. DEFAULTS UPON SENIOR SECURITIES

None.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

ITEM 5. OTHER INFORMATION

Not applicable.

ITEM 6. EXHIBITS

Exhibit Number		Description of Document
31.1	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Executive Officer.
31.2	—	Rule 13a-14(a)/15d-14(a) Certification of Chief Financial Officer.
32.1	—	Section 1350 Certification of Chief Executive Officer.
32.2	—	Section 1350 Certification of Chief Financial Officer.
101	—	The following materials from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended September 30, 2019, formatted in Inline XBRL (Extensible Business Reporting Language): (i) the Consolidated Balance Sheets, (ii) the Consolidated Statements of Income, (iii) the Consolidated Statements of Comprehensive Income, (iv) the Consolidated Statements of Changes in Equity and Redeemable Non-Controlling Interest (v) Consolidated Statements of Cash Flows and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text.
104	—	The cover page from Intercontinental Exchange, Inc.'s Quarterly Report on Form 10-Q for the quarter ended September 30, 2019, formatted in Inline XBRL.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized.

Intercontinental Exchange, Inc.
(Registrant)

Date: October 31, 2019

By: /s/ Scott A. Hill

Scott A. Hill

Chief Financial Officer

(Principal Financial Officer)

CERTIFICATIONS

I, Jeffrey C. Sprecher, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: October 31, 2019

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chairman of the Board and
Chief Executive Officer

CERTIFICATIONS

I, Scott A. Hill, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Intercontinental Exchange, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: October 31, 2019

/s/ Scott A. Hill

Scott A. Hill

Chief Financial Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the "Company") on Form 10-Q for the period ended September 30, 2019 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), I, Jeffrey C. Sprecher, Chairman and Chief Executive Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: October 31, 2019

/s/ Jeffrey C. Sprecher

Jeffrey C. Sprecher
Chairman of the Board and
Chief Executive Officer

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Quarterly Report of Intercontinental Exchange, Inc. (the “Company”) on Form 10-Q for the period ended September 30, 2019 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), I, Scott A. Hill, Chief Financial Officer of the Company, certify, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that, to the best of my knowledge:

- (1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
- (2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: October 31, 2019

/s/ Scott A. Hill

Scott A. Hill

Chief Financial Officer